

CivicWorkOS: Capability-Preserving Allocation of Municipal Work Among Humans, AI Agents and Robots

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2 ABSTRACT

Cities are moving from sensing infrastructure to agentic operation, in which AI agents perform administrative work and robots perform physical work. Allocation research optimizes fit, throughput, and ergonomics; labor research examines displacement; neither prices the developmental practice through which cities form their future practitioners. We present CivicWorkOS, which makes municipal allocation a governed decision over *staffed modes* — pairings of an execution mode with a named roster at declared career stages — rather than a human-versus-machine choice. Civic Automation Debt prices five deferred liabilities, and four constraints bind the objective: a Human Capability Preservation Budget denominated in qualified-practice hours accruing only to practitioners below independent competence, a Capability Access Constraint governing who receives protected practice, a Just Transition Constraint, and an N–1/N–2 resilience reserve. We give the city-wide program, an online Lagrangian rule with bounded between-rebalance violation, a feasibility condition, and an intake requirement for when the budget is unattainable. An exact bridge-inspection instance, reproducible by direct arithmetic from the reported values, prices protected practice at 0.0454 objective units per hour, costs 4.34% of annual objective value (0.33–9.60% across the swept parameter frontier), and yields 3.46 competent-equivalents a year against attrition of 2.88. The same arithmetic produces a result no headcount plan surfaces: machine-assisted supervised practice stretches time to competence by a factor of 3.38, from 1.2 to 4.05 years, so a city sizing intake from certification hours provisions four trainee posts where fourteen are required.

Keywords: human–AI–robot task allocation, municipal automation governance, capability preservation, civic automation debt, workforce skill formation, algorithmic contestability, urban digital twin, constrained optimization

1 INTRODUCTION

Urban automation has changed character. Traditional smart-city systems sensed, connected, and analyzed urban data. Agentic systems additionally plan, decide, negotiate, invoke tools, and act, while autonomous

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robots physically execute those decisions in public space. Choi and Yoon (2025) demonstrate AI-agent-based urban digital twins operating over live municipal data, and Shariatpour and Behzadfar (2026) describe the resulting shift toward “agentic cities” in which urban infrastructures sense, learn, predict, and act.

This creates a distributional question that allocation mechanisms do not yet pose. A bridge inspection may be performed by a licensed inspector, a computer-vision agent, an autonomous robot, or a mixed team; a citizen service request by a public employee, an AI agent, or a human supervised by one; and sanitation, maintenance, transport, logistics, emergency response, and administration decompose the same way. Every such decision has a winner and a loser, and cities make them without any mechanism that records who lost what.

The policy problem is not whether AI and robots will replace municipal workers, but how a city should continuously decide which tasks are automated, augmented, reserved for humans, or performed by hybrid teams, and who performs the work that remains.

We treat that second half as constitutive. Neither automation exposure (Acemoglu and Restrepo, 2020; Eloundou et al., 2024) nor the developmental work that turns a novice into an expert is distributed evenly across occupational class, gender, contract status, or district. A municipality that automates its junior inspection, triage, and casework is not only reducing headcount. It is closing the route by which its next senior practitioners would have been formed, closing it first for whoever was next in line, and doing so invisibly to both a productivity objective and a post-deployment equity audit, because it operates one locally defensible task at a time. It also forms each remaining practitioner more slowly: in the worked allocation of Section 6, machine assistance and shared supervision stretch formation from the 1.2 years certification assumes to 4.05, so a city sizing intake from certification hours provisions four trainee posts where fourteen are required (Figure 3).

1.1 Contributions

Most components adapt established constructions, as Section 2 records; the contribution lies in the decision variable that joins them and in what the joining makes computable.

- A city-scale allocation formalism whose decision variable is a *staffed mode*, a pairing of an execution mode with a roster of named practitioners at declared career stages. This is what makes the distributional questions answerable: mode selection alone cannot express who receives the work.
- *Civic Automation Debt*, pricing the deferred liability created by skill-formation loss, fallback erosion, accountability dilution, vendor dependency, and labor-transition burden. We claim no novelty for the underlying concern, which Bainbridge (1983) stated for the individual operator. The contribution is its aggregation to a municipal workforce, its formal separation from the immediate-outcome terms with which it is routinely confused, and its entry into a decision as a priced quantity with a computable flip threshold.
- A *Human Capability Preservation Budget* in qualified-practice hours that accrues only to practitioners below independent competence, with a stated unit conversion from raw certification hours and an explicit feasibility condition.
- A *capability intake requirement* and the *dilution factor* that drives it, which turn the budget into a hiring number and show that automation slows the formation of each practitioner as well as reducing their number (Section 6.6).
- A *Capability Access Constraint* governing who receives protected practice, so a budget sized from the incumbent workforce does not reproduce its composition for a replacement cycle, together with the equality-law and data-protection analysis that a mechanism allocating work by protected characteristic requires.
- A *Policy Digital Twin* with a rule language, status lattice, and conflict order, and *Adaptive Oversight Zones* with specified transition predicates.
- The city-wide assignment program the coupled constraints require, an online Lagrangian approximation with a bounded between-rebalance violation, and a bridge-inspection allocation carried through end to end, with a sensitivity sweep and two parameter shocks, by exact arithmetic.
- An evaluation protocol specified in full, with ten hypotheses and refutation criteria. It has not been executed, and no claim in this article rests on an outcome of it.

77 Section 2 positions the work and states the research questions; Section 3 defines the framework's layers;
 78 Section 4 gives the program, algorithm, and formal results; Section 5 covers governance and lawfulness;
 79 Section 6 works the bridge-inspection allocation end to end; and Section 7 states what that establishes, the
 80 unexecuted evaluation protocol, and the limitations.

2 RELATED WORK AND RESEARCH GAP

81 This section states our positioning; Appendix S6 gives the survey work by work.

82 2.1 Task Allocation and Workforce Scheduling

83 Capability-based allocation matches task requirements to agent capabilities, so that allocation follows
 84 task-agent fit rather than automation for its own sake (Ranz et al., 2017; Dorri et al., 2018). Later work
 85 added ergonomics (El Makrini et al., 2019; Merlo et al., 2023), human preferences (Noormohammadi-Asl
 86 et al., 2025), and measured trust (Ali et al., 2022), with reviews covering robot fleets (Meseguer Valenzuela
 87 and Blanes Noguera, 2025) and multi-agent reinforcement learning (Hady et al., 2025); the closest city-
 88 scale treatment allocates sensing tasks across a mobile crowd as a graph-structured assignment (Zhao et al.,
 89 2023). All of it optimizes an operational team's performance. None asks whether repeated local decisions
 90 erode the experience that sustains future expertise, or whose expertise erodes first.

91 The capability budget of Section 3.5 belongs to the long operations-research tradition of constrained
 92 training-hour allocation rather than departing from it: nurse rostering carries skill mix, supervision ratios,
 93 and qualification maintenance as hard constraints (Burke et al., 2004), and personnel scheduling includes
 94 cross-training variants (Van den Bergh et al., 2013). Two differences matter. Rostering assumes the work
 95 exists and asks who covers it, whereas our decision is prior — whether a human performs it at all, and
 96 so whether developmental content exists to allocate — and rostering optimizes coverage, whereas our
 97 objective prices the future value of the practice itself.

98 2.2 Automation, Skill Erosion, and Capability Formation

99 The concern at the center of this paper is not new, and its lineage belongs on the record. Automating the
 100 easy parts of a task degrades operator skill and leaves the human unable to intervene when intervention is
 101 required (Bainbridge, 1983); the out-of-the-loop problem (Endsley and Kiris, 1995), the use, misuse, disuse,
 102 and abuse failure modes (Parasuraman and Riley, 1997), and the levels-of-automation scales (Sheridan
 103 and Verplank, 1978; Parasuraman et al., 2000) are long established, and Shneiderman (2020) separates
 104 automation from human control as independent axes. Zuboff's (1988) distinction between automating
 105 and informing work is the distinction between a task that displaces a practitioner and one that develops
 106 them. Skill-decay meta-analyses (Arthur et al., 1998; Tatel and Ackerman, 2025) supply the instruments to
 107 calibrate D^{skill} against observed practice gaps.

108 CivicWorkOS therefore does not claim to have discovered skill erosion. That literature works at the scale
 109 of the individual operator, whereas our question — whether a city's expertise pipeline is drawn down across
 110 thousands of independent decisions — is municipal, and it warrants treating fallback capacity separately
 111 from present competence, as D^{fall} does. The debt framing has a source too: technical debt as the deferred
 112 cost of an expedient choice (Cunningham, 1992; Sculley et al., 2015), here a liability settled later, by
 113 someone else, invisible on the ledger that authorized it. Rashidi (2026) argues that AI is the first technology
 114 able to automate the entry-level cognitive work through which professional judgment is formed.

115 2.3 Fair Division and Algorithmic Due Process

116 Protected practice is a scarce, unequally distributed good, placing the Capability Access Constraint inside
 117 the fair-division literature (Bouveret et al., 2016; Caragiannis et al., 2019). We use a share constraint against
 118 a policy-set target rather than an envy-based criterion because a municipal panel must defend a distribution
 119 against a statutory equality duty rather than against agents' own valuations, and because a Nash-welfare
 120 objective would depend on worker preferences no city records and workers would report strategically. The
 121 cost is that our constraint offers no envy guarantee.

122 A framework that prices civic costs must also show its working, or the pricing is unfalsifiable in the way
 123 opaque public-sector scoring is (Eubanks, 2018). Technological due process requires notice, a record, and

a route to challenge (Citron, 2008; Citron and Pasquale, 2014), and a right to human intervention is empty unless a human can alter the outcome (Almada, 2019; Vaccaro et al., 2019). Of the contestability features Alfrink et al. (2023) synthesize, Section 5.2 implements the evidentiary record, the intervention request, and challenge to the rules themselves, and adds a systemic trigger because formally fair prioritization can still produce outcomes experienced as unjust (Moon and Guha, 2026). Article 22 GDPR (European Parliament and Council of the European Union, 2016) sets the EU floor.

2.4 Algorithmic Management, Labor, and Who Bears the Cost

AI is becoming a collaborator (Senjic et al., 2026) and a manager: algorithmic management already allocates work, issues instructions, and monitors performance (Milanez et al., 2025), with gains in consistency set against autonomy, privacy, accountability, and discrimination concerns (OECD, 2026) that persist even where automation performs well (Wischnewski et al., 2023). Municipalities differ because public services are constituted by the discretionary judgment of street-level bureaucrats (Lipsky, 2010), so removing discretion changes what the service is. Ethical frameworks (Callari et al., 2024), the EU AI Act's oversight requirement (European Parliament and Council of the European Union, 2024), and the ISO 10218 series (International Organization for Standardization, 2025a,b) specify principles or physical constraints, not a mechanism that turns them into real-time decisions about who performs a task.

On employment, the task-based framework (Autor et al., 2003) underpins the argument that displacement is overstated when complementarities are ignored (Autor, 2015), and occupation-level exposure estimates (Frey and Osborne, 2017) fall roughly fivefold when re-measured at task level (Arntz et al., 2016); only task redesign is actionable by an allocation mechanism. Language models widen exposure (Eloundou et al., 2024), robot adoption lowers employment and wages for lower-skilled workers (Acemoglu and Restrepo, 2020), and exposure falls unevenly, on platform workers outside municipal protections (Wood et al., 2019; De Stefano, 2016) and on the poorest residents (Eubanks, 2018). The ILO just-transition guidelines specify what an institutional response must contain (International Labour Organization, 2015), alongside OECD policy (OECD, 2026), employer reskilling forecasts (World Economic Forum, 2025), and Arab-region projections that Section 5.4 takes up (International Labour Organization and United Nations Economic and Social Commission for Western Asia, 2026).

We use *capability* in the sense of Sen (1999) and Nussbaum (2011), the real freedom to do and to be that a developmental task confers and an automated one withholds, and the budget takes the institutional form Ostrom (1990) identifies for depletable shared resources: monitored, bounded, locally governed appropriation rules. None of these supplies a procedure a city can run on Tuesday morning, for one task, pricing exposure before the allocation rather than compensating for it afterwards.

2.5 Agentic-City Governance and Urban Digital Twins

Public-space robot policy concentrates on safety and privacy over wider wellbeing (Mintrom et al., 2026); co-creation and transparent civic platforms raise measured trust in municipal AI (Phothong et al., 2026; Nechesov and Ruponen, 2024), and the workforce's own digital readiness conditions reform (Aldhi et al., 2025). City digital twins already assist policy decisions (Temple et al., 2024), and Diaz-Sarachaga (2025) supplies criteria for whether a deployed twin is well governed, the closest antecedent to our Policy Digital Twin. We ask the converse: whether governance can be represented inside the twin as an executable constraint, carrying the social dimensions twins usually omit (Ben-Daya et al., 2026; Qanazi et al., 2025). Resilience is a related but distinct concept, used inconsistently across the infrastructure literature (Mentges et al., 2023); response networks recover faster when post-failure reassignment is planned in advance (Lian et al., 2025).

2.6 Research Gap and Research Questions

Table 1 positions CivicWorkOS against the closest published frameworks on the dimensions that matter for a distributional claim.

Five gaps follow. They are gaps of integration more than of method: each construction we use has a precedent above, and what is missing is their joining at municipal scale. First, allocation models and policy frameworks exist separately, and we know of no city-scale mechanism that makes policy an active constraint of allocation. Second, the single-operator mechanism of skill erosion has not been aggregated to a municipal workforce and priced.

Table 1. Positioning of CivicWorkOS against the closest published allocation and governance frameworks. Columns: task–agent fit; multi-objective objective; deferred cost priced; mechanism decides who receives the work; policy enters as an executable constraint; contestability procedure specified; primary scale of application. ✓ present and specified; ▲ partially present or asserted without mechanism; ✗ absent.

Framework	Fit	Multi-obj.	Debt priced	Who works	Policy	Contest.	Scale
Ranz et al. (2017)	✓	✗	✗	✗	✗	✗	Workcell
El Makrini et al. (2019)	✓	▲	✗	✗	✗	✗	Workcell
Merlo et al. (2023)	✓	✓	✗	✗	✗	✗	Workcell
Ali et al. (2022)	✓	▲	✗	✗	✗	✗	Team
Zhao et al. (2023)	✓	✓	✗	✗	✗	✗	City sensing
Nurse rostering (Burke et al., 2004)	✓	✓	▲	✓	▲	✗	Department
Diaz-Sarachaga (2025)	✗	✗	✗	✗	▲	▲	City
Alfrink et al. (2023)	✗	✗	✗	✗	▲	✓	System
CivicWorkOS	✓	✓	✓	✓	✓	✓	City

175 Third, and most consequentially, no allocation mechanism we know of asks who receives the work that is
 176 preserved. A preservation rule calibrated on the current workforce reproduces its composition — gender
 177 balance, credentialing routes, neighborhoods, ratio of permanent to contracted staff — under the honorable
 178 name of preserving expertise, converting a protective mechanism into an exclusionary one.

179 Fourth, allocation needs a measure of whether enough human capability remains to recover essential
 180 services when AI, platforms, or robots fail. Fifth, automation rules are static compliance checks, whereas
 181 the acceptable degree of automation varies with sector, risk, labor conditions, and observed performance.
 182 These gaps motivate seven research questions, addressed in Sections 3 to 7.

183 **RQ1**How can urban tasks be allocated among humans, AI agents, robots, and hybrid teams on both
 184 operational and civic objectives?

185 **RQ2**Can Civic Automation Debt price long-term capability losses without double-counting immediate
 186 outcomes?

187 **RQ3**Can a capability budget sustain expertise pipelines without fixed quotas, and at what cost?

188 **RQ4**Who receives protected developmental work, and can an access constraint stop preservation
 189 reproducing the incumbent workforce?

190 **RQ5**How can governance requirements become machine-checkable constraints with an operable
 191 contestability procedure?

192 **RQ6**Can a resilience reserve preserve critical services under automation failure?

193 **RQ7**How should oversight zones respond to performance, incidents, workforce capability, and feedback?

3 THE CIVICWORKOS FRAMEWORK

194 3.1 Problem Statement, Staffed Modes, and Civic Automation Debt

195 Consider a city containing urban tasks $\mathcal{T} = \{T_1, \dots, T_n\}$ and a heterogeneous workforce of humans (H),
 196 software AI agents (A), physical robots (R), and their hybrid combinations, giving the execution-mode set
 197 $\mathcal{M}$ defined in equation (1):

$$\mathcal{M} = \{H, A, R, H+A, H+R, A+R, H+A+R\}. \quad (1)$$

198 Each task belongs to one critical service $\sigma(i) \in \mathcal{S}$ and one capability domain $\kappa(i) \in \mathcal{K}$, both taken from
 199 the municipal service catalog.

200 Practitioners, Career Stage, and Staffed Modes

201 Selecting a mode alone cannot express who receives the work, so we make the human staffing explicit.
 202 Let $\mathcal{W}_k$ be the practitioners qualified in domain k , each carrying an accumulated qualified-practice total
 203 $c_w(t)$ and facing a competence threshold h_k , which gives the *career stage* of equation (2):

$$\text{stage}_k(w, t) = \begin{cases} \text{developing}, & c_w(t) < h_k, \\ \text{competent}, & c_w(t) \geq h_k. \end{cases} \quad (2)$$

204 Only competent practitioners may hold roles carrying statutory sign-off. $N_k(t)$ counts competent
 205 practitioners in domain k and $N_k^{dev}(t)$ the developing ones.

206 A mode m specifies $n_m \geq 0$ human slots. A *roster* a assigns those slots to distinct available practitioners
 207 with a work split $\omega_{a,w} \geq 0$, $\sum_{w \in a} \omega_{a,w} = 1$; modes with no human slot admit only the null roster. The
 208 pair (m, a) is a *staffed mode*, and the decision variable is defined in equation (3):

$$x_{i,m,a} \in \{0, 1\}, \quad \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} = 1 \quad \forall i, \quad (3)$$

209 where $\mathcal{A}_{i,m}$ is the set of feasible rosters. The *developmental eligibility* of a roster, equation (4),

$$\psi_a(t) = \sum_{w \in a} \omega_{a,w} \mathbb{I}[\text{stage}_{\kappa(i)}(w, t) = \text{developing}] \in [0, 1], \quad (4)$$

210 is the share of the human work performed by practitioners still being formed, and the *group share* of
 211 equation (5),

$$\pi_{a,g} = \sum_{w \in a} \omega_{a,w} \mathbb{I}[w \in g], \quad \sum_{g \in \mathcal{G}} \pi_{a,g} = 1, \quad (5)$$

212 is well defined for a hybrid team drawn from several groups and makes the protected-practice shares of
 213 Section 3.6 sum to one by construction. Worker groups $\mathcal{G}$ are a coarsening of the protected attributes
 214 declared by the panel of Section 5.1 subject to a minimum cell size, not their Cartesian product
 215 (Appendix S7.1).

216 Civic Automation Debt

217 An allocation chosen on time, cost, accuracy, or safety generates externalities invisible at the task level:
 218 repeated automation removes entry-level learning, concentrates knowledge in proprietary systems, erodes
 219 fallback capacity, shifts accountability, and imposes transition costs on identifiable workers. We define
 220 *Civic Automation Debt* (CAD) as the accumulated future liability created when automation removes
 221 human practice, concentrates knowledge, reduces fallback capacity, increases dependency, or displaces
 222 workers faster than the city develops alternatives. For task T_i under staffed mode (m, a) it is quantified by
 223 equation (6):

$$CAD_{i,m,a} = \alpha D_{i,m,a}^{skill} + \beta D_{i,m}^{fall} + \gamma D_{i,m}^{acct} + \delta D_{i,m}^{dep} + \epsilon D_{i,m}^{trans}, \quad (6)$$

224 where D^{skill} is skill-formation loss, D^{fall} loss of human fallback capability, D^{acct} accountability dilution,
 225 D^{dep} technological dependency, and D^{trans} labor-transition burden, with weights summing to one
 226 (Section 4.3). CAD here never denotes computer-aided design.

227 If these components restated quantities the objective already contains, the model would double-count.
 228 We separate them by measurement domain. The terms of (25) are *flows on task T_i 's own execution window*,
 229 whereas the CAD components are *changes in municipal stocks* integrated over the horizon in which they
 230 settle: the pipeline, fallback capacity, unaccountable decision steps, vendor concentration, and workers
 231 without a reskilling pathway. Equity appears twice but as two objects: Eq^{srv} is the equity of *service to*
 232 *residents* in this window, D^{trans} the burden on *workers* over the settlement horizon. Developmental content
 233 reaches the decision through four distinct channels, which Appendix S7.7 separates and Section 4.7 weighs.

Each D term is normalized to $[0, 1]$ and estimable from signals a city already tracks (Appendix S7.2). Skill-formation loss is the one that carries the framework, equation (7):

$$D_{i,m,a}^{skill} = 1 - \phi_m \psi_a, \quad (7)$$

where ϕ_m is the share of the task's developmental content a human experiences under mode m (Section 3.5). Both factors are needed: a task performed entirely by machine forms nobody, and neither does one performed by an already-competent practitioner alone. An allocation that maximizes current productivity while continuously increasing CAD draws down stocks no term in its own objective replenishes.

3.2 Layer 1: Urban Task Digital Twin

Every service is decomposed into tasks carrying a requirement vector and a duration, equation (8):

$$T_i = \{cog_i, phy_i, emp_i, risk_i, auth_i, priv_i, urg_i, learn_i, crit_i; d_i\}, \quad (8)$$

covering cognitive, physical, and social demand, safety risk, legal-authority requirement, privacy sensitivity, urgency, human learning value, criticality, and execution time in hours, each normalized to $[0, 1]$ (Appendix S7.3). The task's *developmental content*, equation (9),

$$\ell_i = learn_i \cdot d_i \quad [\text{qualified-practice hours}], \quad (9)$$

is the unit in which the capability budget is denominated. The ISO 23247 reference architecture (Shao et al., 2023) covers the sensing half of (8) but not $learn_i$ or $crit_i$, which are municipal policy judgments rather than measurements.

3.3 Layer 2: Civic Capability Ledger

The Civic Capability Ledger records, for each domain k , the state of equation (10):

$$L_k(t) = \{H_k(t), A_k(t), R_k(t), F_k(t), E_k(t), \{E_{k,g}(t)\}_{g \in \mathcal{G}}, \{c_w(t)\}_{w \in \mathcal{W}_k}\}, \quad (10)$$

comprising available human, AI, and robotic capability, fallback capacity F_k , the expertise pipeline E_k and its decomposition by worker group, without which the access constraint cannot be evaluated, and the per-practitioner totals that (2) reads. Fallback capacity is defined in equation (11):

$$F_k(t) = \sum_{w \in \mathcal{W}_k} \zeta_w(t) \mathbb{K}[\text{stage}_k(w, t) = \text{competent}], \quad (11)$$

the qualified hours deployable under surge within a service's recovery window, with $\zeta_w(t)$ practitioner w 's surge-deployable hours net of committed work. Because it is conditioned on competence, F_k decays on the same clock as the pipeline.

3.4 Layer 3: Policy Digital Twin

The Policy Digital Twin converts laws, rules, collective agreements, and local priorities into machine-checkable constraints. A policy rule is the tuple of equation (12):

$$\rho = \langle \text{scope}, \text{guard}, \text{status}, \text{authority}, \text{validity}, \text{provenance} \rangle, \quad (12)$$

where the scope is a conjunction of catalog predicates, the guard a decidable predicate over the task vector, roster, and ledger state, the authority ranks the source from statute down to operational guidance, the validity is a time interval, and the provenance names the instrument and the panel decision that encoded it. Statuses form the totally ordered lattice of equation (13),

$$\text{allow} \prec \text{allow-with-oversight} \prec \text{restrict} \prec \text{prohibit}, \quad (13)$$

and the twin returns the join over every firing rule, equation (14):

$$\mathcal{P}(T_i, m, a, t) = \bigsqcup_{\rho \in \text{fires}(T_i, m, a, t)} \text{status}(\rho) \in \{\text{allow}, \text{allow-with-oversight}, \text{restrict}, \text{prohibit}\}. \quad (14)$$

The most restrictive applicable rule therefore wins, the correct default for a public body, and a lower-ranked rule can never relax a higher-ranked one. The two intermediate statuses are executable: *allow-with-oversight* binds a named reviewer to the staffed mode and adds the reviewer's time and retained judgment to its terms, while *restrict* narrows $\mathcal{A}_{i,m}$ to rosters meeting a minimum human share or stage requirement, which is how a supervision ratio enters the model. Appendix S7.4 gives the conflict order, the consistency check, and a worked encoding of Article 14 of the EU AI Act (European Parliament and Council of the European Union, 2024); Section 5.2 makes the twin's decisions contestable.

3.5 Layer 4: Human Capability Preservation Budget

The Human Capability Preservation Budget (HCPB) reserves a minimum volume of developmental human practice, denominated throughout in *qualified-practice hours per budget period* ΔT , taken as one year, because a budget whose units do not close cannot be audited. The *human developmental share* of mode m is given in equation (15):

$$\phi_m \in [0, 1], \quad \phi_H = 1, \quad \phi_A = \phi_R = \phi_{A+R} = 0, \quad (15)$$

with hybrid values set per domain from an assessment of how much of the task's judgment a human still exercises. The HCPB constraint is equation (16):

$$\Lambda_k(\Delta T) = \sum_{i \in \mathcal{T}_k(\Delta T)} \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} \ell_i \phi_m \psi_a \geq B_k(t), \quad (16)$$

both sides in hours delivered to practitioners below competence. The factor ψ_a is the substantive repair: without it a domain that gives every task to its incumbents satisfies a budget sized from novice formation while forming nobody. The budget is estimated in equation (17):

$$B_k(t) = \max\left(B_k^{\min}, \eta_k N_k(t) r_k(t) h_k\right), \quad (17)$$

where $r_k(t)$ is the fraction of competent practitioners lost per period, h_k the qualified-practice hours that bring one developing practitioner to competence, η_k a policy-set replacement factor ($\eta_k > 1$ builds margin, $\eta_k < 1$ deliberately lets a domain shrink), and $B_k^{\min}$ an hours floor. Certification schemes count *raw supervised clock hours* h_k^{raw} under solo unassisted practice, which accrues the mean learning value $\overline{\text{learn}}_k$ per clock hour, so equation (18) converts between them:

$$h_k = \overline{\text{learn}}_k \cdot h_k^{\text{raw}}. \quad (18)$$

A practitioner on machine-assisted supervised tasks accrues less than $\overline{\text{learn}}_k$ per clock hour, which is why realized formation takes longer than certification assumes (Section 6.6). The budget is not a jobs quota: it constrains a volume of practice rather than a number of posts, so a task may go to a developing human supported by an AI agent even where AI could perform it alone. In Ostrom's terms (1990) it is an appropriation rule over a renewable common-pool resource, bounded, monitored, and revised by the resource's own users through Section 5.1.

3.6 Layer 4b: Capability Access and Just Transition

Equation (16) decides *how much* developmental work is protected but not *who* performs it, and left alone it has a predictable answer: because developing practitioners are recruited through the pipelines that produced the incumbents, a composition-blind budget reproduces the incumbent composition for a full replacement cycle under the name of preserving capability. We therefore separate the budget's *size*, derived from attrition, from its *distribution*, which must be set deliberately. The protected practice share of group g

is defined in equation (19):

$$\Pi_{k,g} = \frac{\sum_{i \in \mathcal{T}_k(\Delta T)} \sum_m \sum_a x_{i,m,a} \ell_i \phi_m \psi_a \pi_{a,g}^{dev}}{\sum_{i \in \mathcal{T}_k(\Delta T)} \sum_m \sum_a x_{i,m,a} \ell_i \phi_m \psi_a} \quad (19)$$

where $\pi_{a,g}^{dev}$ is the group share (5) restricted to the roster's developing members, so that $\sum_g \Pi_{k,g} = 1$. The *Capability Access Constraint* is equation (20):

$$\Pi_{k,g} \geq \theta_{k,g} - \varepsilon_k \quad \forall g \in \mathcal{G}, \quad (20)$$

with target composition $\theta_{k,g}$ and tolerance ε_k . The critical design choice is that $\theta_{k,g}$ is *not* the incumbent composition. It is set by the panel of Section 5.1 from the qualified local labor pool, a statutory equality duty, or an explicit equity target, and published with the weight vector; setting it to the incumbent share remains permitted, but is then a recorded decision to freeze composition rather than an unnoticed property of the arithmetic. Feasibility requires $\sum_g (\theta_{k,g} - \varepsilon_k) \leq 1$, which the twin checks on publication. Section 5.3 addresses whether such a target may lawfully be set.

The second distributional constraint concerns those who lose work. With $\Delta_g(t, t + \Delta T)$ the net reduction in group g 's allocated hours as a fraction of its baseline, the *Just Transition Constraint* is equation (21):

$$\Delta_g(t, t + \Delta T) \leq \tau_g \quad \forall g \in \mathcal{G}, \quad (21)$$

a ceiling on the pace of displacement for any single group, following the ILO guidelines (International Labour Organization, 2015), which stops the framework concentrating displacement on the group with least bargaining power. Reskilling coverage is a municipal budget fact rather than a function of $\mathbf{x}$, so it is enforced in the governance layer as a gating predicate rather than in the program (Appendix S7.5). Together, (20) and (21) do not resolve who wins and who loses, but make the answer explicit, versioned, auditable, and contestable.

3.7 Layer 5: Reversibility and Resilience Reserve

For each critical service s the city maintains a minimum recoverable capacity that does not depend on a single automation stack. Total service capacity comprises the automation-dependent components $\mathcal{C}_s^{aut}$ (an AI agent pool, a robot fleet class, a vendor platform) and the human contribution of the domains serving s , equation (22):

$$C_s(t) = \sum_{c \in \mathcal{C}_s^{aut}} C_{s,c}(t) + \sum_{k \in \mathcal{K}_s} \xi_{s,k} F_k(t), \quad (22)$$

with $\xi_{s,k}$ the share of domain k 's surge-deployable hours that s may draw. Surviving capacity after losing the n largest automation-dependent components $\mathcal{C}_s^{(n)}$ is equation (23):

$$Res_s^{(n)}(t) = C_s(t) - \sum_{c \in \mathcal{C}_s^{(n)}(t)} C_{s,c}(t), \quad (23)$$

so what survives an AI-service outage or a fleet compromise is exactly the human capacity in the second term of (22). The *Reversibility and Resilience Reserve* constraint is equation (24):

$$Res_s^{(n_s)}(t) \geq \rho_s, \quad \rho_s = \kappa_s D_s^{peak}, \quad (24)$$

with contingency order $n_s \in \{1, 2\}$, peak demand D_s^{peak} , and $\kappa_s \in (0, 1]$ the share of that peak which must remain servable. This is the N-1 and N-2 criterion of reliability engineering (Jain et al., 2024; Ouyang, 2014) applied to a workforce of humans, AI agents, and robots. Because F_k decays when a domain stops forming practitioners, the resilience and capability constraints are coupled.

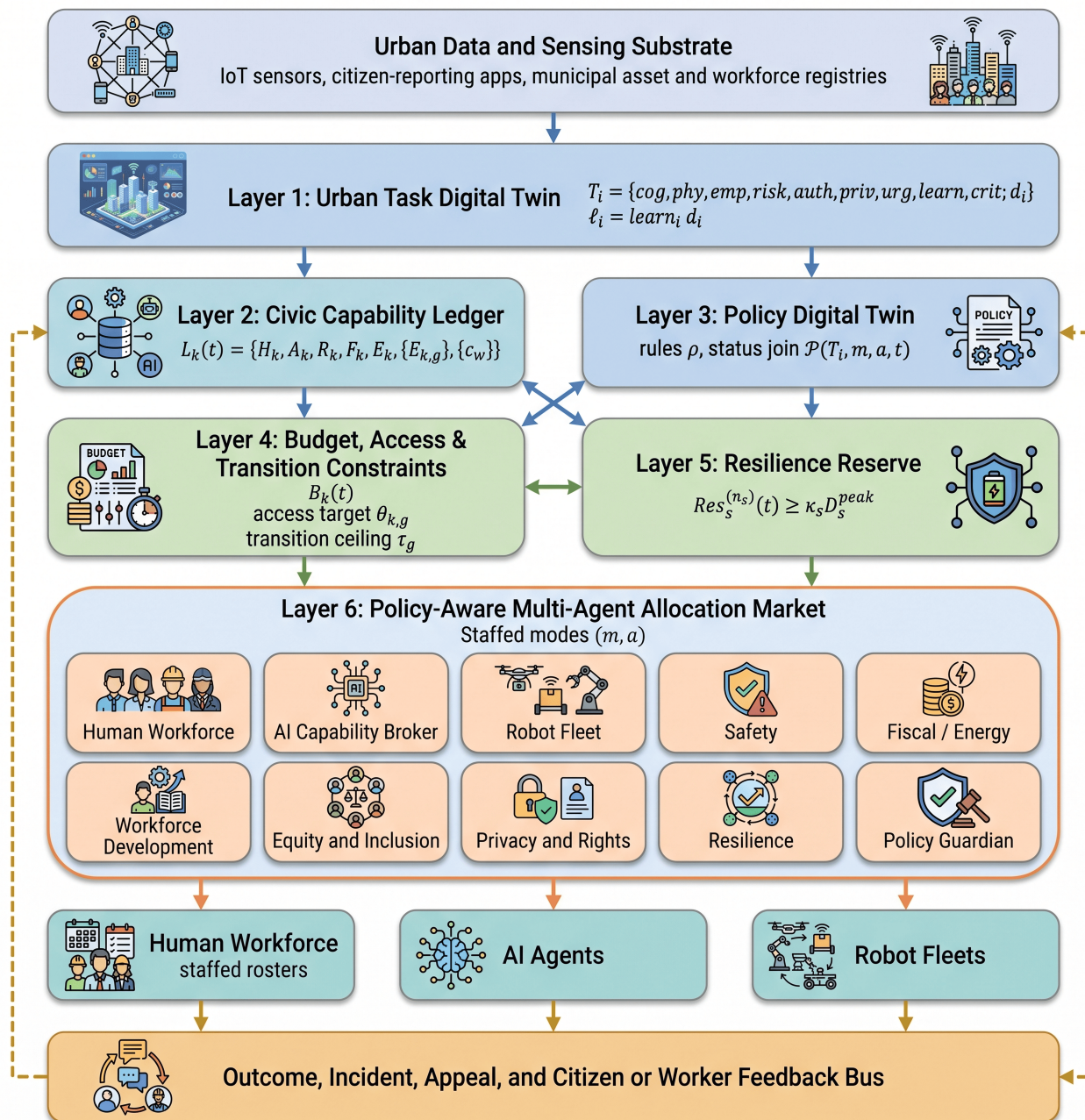


Figure 1. Reference architecture of CivicWorkOS, read top to bottom. Solid arrows are the forward allocation path from sensed urban data through the governance layers to execution; the ten agents are drawn inside the market because they constitute it. The Ledger supplies fallback capacity F_k of Equation (11) to the reserve through Equation (22), and per-practitioner practice totals c_w to the career-stage test of Equation (2). The Policy Twin supplies the replacement factor η_k , the access target $\theta_{k,g}$, and the roster guards that restrict $\mathcal{A}_{i,m}$. Dashed arrows are the outcome-feedback path, which updates both governance services.

3.8 Layer 6: Policy-Aware Multi-Agent Allocation Market

Ten specialized agents price candidate staffed modes: five *capability agents* estimate the flow terms and five *civic agents* the stock changes, among them the Policy Guardian Agent, which executes (14) and holds a veto (Figure 1). The market is a single-round sealed-bid call, a structured elicitation mechanism rather

than a strategic game, and we make no equilibrium claim for it. Appendix S7.8 lists the agents and the protocol, and Section 7.6 records the principal–agent exposure the design leaves open.

4 ARCHITECTURE, OPTIMIZATION PROGRAM, AND ALGORITHM

4.1 Reference Architecture

Figure 1 realizes the layers of Section 3 as an event-driven system in which a single feedback bus returns outcomes, incidents, and appeals from all three execution channels to both governance services. Robot execution sits behind fleet middleware enforcing ISO 10218-2 (International Organization for Standardization, 2025b), with the robots conforming to ISO 10218-1 (International Organization for Standardization, 2025a), so a policy-level error cannot defeat a hardware-level safety interlock.

4.2 Runtime Allocation Workflow

Figure 2 shows the per-task control flow that Algorithm 1 formalizes. If policy filtering or roster pruning empties the candidate set, the task routes to Zone Z4 human-reserved execution and the event is counted, because its rate is the operational price of the constraint system.

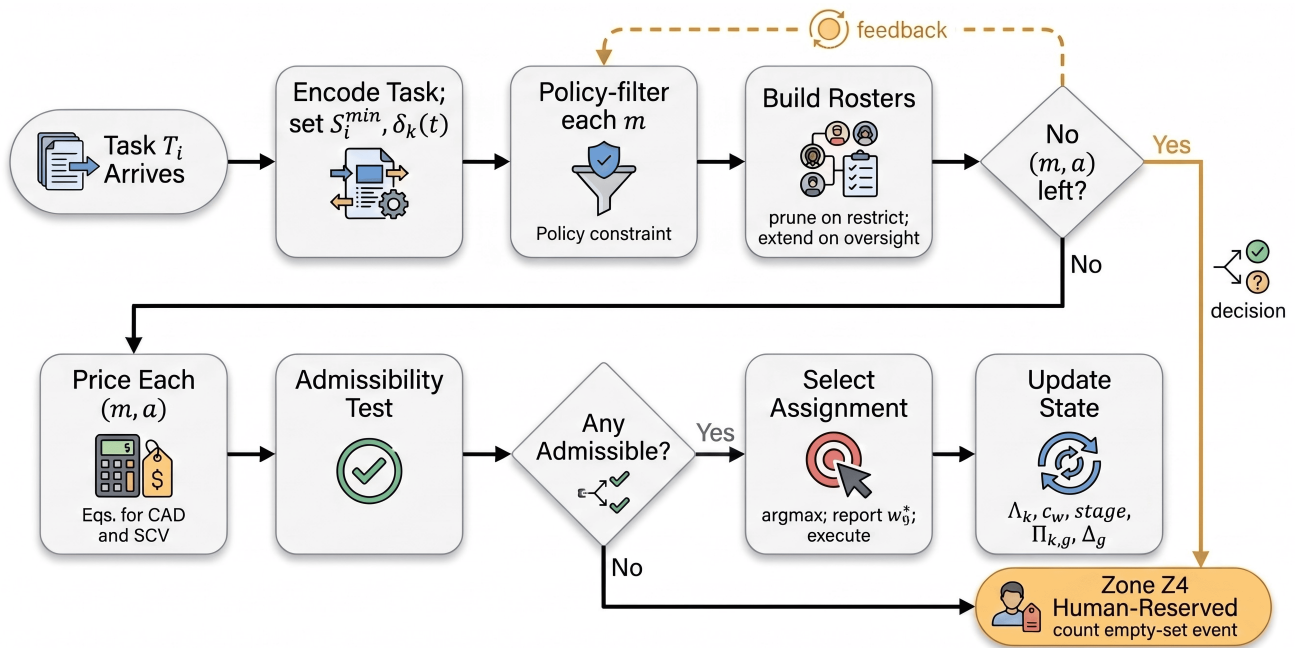


Figure 2. Runtime allocation workflow, matching Algorithm 1. The upper row encodes the task, sets the safety floor and the catch-up increment, policy-filters modes, and builds the roster set, pruning it where the twin returns *restrict* and extending it with a named reviewer where it returns *allow-with-oversight*. The lower row prices the surviving staffed modes, applies the admissibility test (29), and either falls back to Zone Z4 or selects and executes. The dashed edge returns outcomes to the Policy Digital Twin, which is what makes the zone-transition predicates (Section 4.8) evaluable.

4.3 Sustainable Civic Value (SCV)

For each task T_i and staffed mode (m, a) , the *Sustainable Civic Value* is given by equation (25):

$$SCV_{i,m,a} = w_1 Q_{i,m,a} + w_2 S_{i,m,a} + w_3 P_{i,m,a} + w_4 Eq_{i,m,a}^{srv} + w_5 Tr_{i,m,a} - w_6 Cost_{i,m,a} - w_7 En_{i,m} - w_8 Pr_{i,m} - w_9 CAD_{i,m,a}, \quad (25)$$

where Q is service quality, S safety, P productivity, Eq^{srv} the equity of service delivery across resident groups, Tr agent trust, $Cost$ operating cost, En energy use, Pr privacy risk, and CAD the debt of (6). The first eight terms are flows on the task's execution window and the ninth a change in municipal stocks; six depend on the roster as well as the mode.

A linear weighting trades flexibility for interpretability relative to Pareto-front methods (Osika et al., 2023; Zheng and Rasheed, 2026), and recovers only solutions on the convex hull of the attainable set, which bounds the panel's authority (Section 5.1). All nine terms are rescaled to $[0, 1]$: the flow terms against a trailing twelve-month window, and the debt components against externally fixed denominators, since against a rolling window a decaying pipeline would keep scoring mid-range (Appendix S8.2).

Default Weights and What They Actually Carry

The illustrative weights are $w_1=0.15$ (quality), $w_2=0.18$ (safety), $w_3=0.14$ (productivity), $w_4=0.10$ (service equity), $w_5=0.05$ (trust), $w_6=0.08$ (cost), $w_7=0.03$ (energy), $w_8=0.05$ (privacy risk), and $w_9=0.22$ (CAD), with debt weights $\alpha=0.28$, $\beta=0.22$, $\gamma=0.18$, $\delta=0.12$, and $\epsilon=0.20$. They are design defaults for the panel to revise, not fitted estimates; Table S3 assigns a default to every remaining parameter.

Table 2 reports the *effective* weight each construct carries once the weights are composed, because a weight vector can assert a priority the arithmetic withholds. The deferred block carries 0.22, more than productivity at 0.14, but the largest single construct is safety at 0.180: CAD carries the largest weight as a block and as no component. Labor-transition burden carries only 0.044, deliberately: an objective weight can be outvoted, so transition is also carried by the hard constraint (21), and a city that judges displacement unacceptable lowers τ_g rather than raising $w_9\epsilon$. Proposition 3 shows which lever is deciding.

Table 2. Effective weight of each construct in the CivicWorkOS default configuration, obtained by composing $w_9 = 0.22$ with the debt weights. The deferred block is the largest single block of the objective; safety is the largest single construct. Column sums to 1.000.

Construct	Enters via	Effective weight
Safety	w_2	0.180
Service quality	w_1	0.150
Productivity	w_3	0.140
Service equity (residents)	w_4	0.100
Operating cost	w_6	0.080
Skill formation	$w_9\alpha$	0.062
Trust	w_5	0.050
Privacy risk	w_8	0.050
Fallback capacity	$w_9\beta$	0.048
Labor-transition burden (workers)	$w_9\epsilon$	0.044
Accountability	$w_9\gamma$	0.040
Energy	w_7	0.030
Vendor dependency	$w_9\delta$	0.026
<i>Deferred block total</i>	w_9	<i>0.220</i>

4.4 The City-Wide Allocation Program

The constraints of Sections 3.5 through 3.7 couple decisions across tasks, so over a rebalancing window CivicWorkOS solves program (26):

$$\max_{\mathbf{x}, \varsigma} \sum_{i \in \mathcal{T}(\Delta T)} \sum_{m \in \mathcal{M}} \sum_{a \in \mathcal{A}_{i,m}} x_{i,m,a} SCV_{i,m,a} - M \sum_{k,g} \varsigma_{k,g} \quad (26a)$$

$$\text{s.t.} \quad \sum_m \sum_a x_{i,m,a} = 1, \quad \forall i, \quad (26b)$$

$$x_{i,m,a} = 0 \text{ if } \mathcal{P}(T_i, m, a, t) = \text{prohibit}, \quad (26c)$$

$$x_{i,m,a} = 0 \text{ if } S_{i,m,a} < S_i^{\min}, \quad (26d)$$

$$\sum_{i \in \mathcal{T}_k} \sum_m \sum_a x_{i,m,a} \ell_i \phi_m \psi_a \geq B_k(t), \quad \forall k, \quad (26e)$$

$$\Pi_{k,g}(\mathbf{x}) + \varsigma_{k,g} \geq \theta_{k,g} - \varepsilon_k, \quad \varsigma_{k,g} \geq 0, \quad \forall k, g, \quad (26f)$$

$$\Delta_g(\mathbf{x}) \leq \tau_g, \quad \forall g, \quad (26g)$$

$$Res_s^{(n_s)}(\mathbf{x}) \geq \kappa_s D_s^{\text{peak}}, \quad \forall s, \quad (26h)$$

$$\sum_i \sum_m \sum_a x_{i,m,a} u_{i,m,a,r} \leq U_r, \quad \forall r, \quad (26i)$$

$$x_{i,m,a} \in \{0, 1\}. \quad (26j)$$

Here $u_{i,m,a,r}$ is the consumption of resource r (human hours by grade and stage, fleet hours, inference budget) and U_r its availability. The access constraint carries an elastic slack penalized at a large M , so an unattainable target yields a published shortfall rather than infeasibility and never compels an individual assignment, which Section 5.3 shows to be a legal requirement. The program is a binary multi-dimensional assignment problem with side constraints, NP-hard in general, and is therefore solved periodically over a window; this article makes no scalability claim.

4.5 Online Allocation, Duals, and the Admissibility Test

Between rebalances, tasks must be decided on arrival, so CivicWorkOS approximates (26) online by Lagrangian relaxation of all five coupling constraints, with dual prices $\lambda_k, \mu_s, \nu_{k,g}, \varrho_g, v_r \geq 0$ for (26e), (26h), (26f), (26g), and (26i) from the most recent solution. The online score is equation (27):

$$\begin{aligned} \widetilde{SCV}_{i,m,a} = & SCV_{i,m,a} + \lambda_{\kappa(i)} \ell_i \phi_m \psi_a + \mu_{\sigma(i)} \Delta_{\sigma(i)}^{\text{res}}(m, a) + \sum_g \nu_{\kappa(i),g} \ell_i \phi_m \psi_a \pi_{a,g}^{\text{dev}} \\ & - \sum_g \varrho_g \delta_g^{\text{disp}}(i, m, a) - \sum_r v_r u_{i,m,a,r}. \end{aligned} \quad (27)$$

Because one task does not change a fleet's capacity, the resilience term is counterfactual and scaled by the task's share of service demand, equation (28):

$$\Delta_s^{\text{res}}(m, a) = \frac{d_i}{D_s^{\text{tot}}(\Delta T)} \left[Res_s^{(n_s)}(t \mid (m, a) \text{ generalized across } s) - Res_s^{(n_s)}(t) \right], \quad (28)$$

where $D_s^{\text{tot}}(\Delta T)$ is the service's total task-hours in the window. The duals come from the root LP relaxation, the standard construction for Lagrangian decomposition, so (27) approximates the relaxed program with error bounded by the integrality gap the solver reports (Appendix S8.3).

The budget and reserve constraints concern the state of a domain, not of a staffed mode, so testing them inside a loop over (m, a) removes every candidate or none, and a city below threshold would fail hardest in the state the framework exists to prevent. We therefore test the post-execution state and add a feasibility-restoration branch. Staffed mode (m, a) is *admissible* for task i at time t if it satisfies

388 equation (29):

$$\begin{aligned}
 &\mathcal{P}(T_i, m, a, t) \neq \text{prohibit} \quad \text{and} \quad S_{i,m,a} \geq S_i^{\min}, \quad \text{and} \\
 &\left[\text{Res}_s^{(n_s)}(t^+ | m, a) \geq \rho_s \quad \text{or} \quad (\text{Res}_s^{(n_s)}(t) < \rho_s \wedge \Delta_s^{\text{res}}(m, a) \geq 0) \right], \quad \text{and} \\
 &\left[\Lambda_k(t^+ | m, a) \geq \bar{B}_k(t) \quad \text{or} \quad \ell_i \phi_m \psi_a \geq \delta_k(t) \right], \quad \text{and} \\
 &\Delta_g(t^+ | m, a) \leq \tau_g \quad \forall g, \quad \text{and} \quad u_{i,m,a,r} \leq U_r - U_r^{\text{used}}(t) \quad \forall r,
 \end{aligned} \tag{29}$$

389 with $s = \sigma(i)$, $k = \kappa(i)$, where $\Lambda_k(t)$ is the practice accrued in domain k so far in the current budget
 390 period, $\bar{B}_k(t) = B_k(t) \cdot t / \Delta T$ the pro-rated budget trajectory, and the catch-up threshold $\delta_k(t)$ is defined
 391 in equation (30):

$$\delta_k(t) = \frac{\max(0, B_k(t) - \Lambda_k(t))}{n_k^{\text{rem}}(t)} \tag{30}$$

392 the per-task increment that closes the remaining gap over the $n_k^{\text{rem}}(t)$ tasks still expected in the period.
 393 Admitting any mode with $\phi_m \psi_a > 0$ would let a lagging domain stay behind indefinitely; requiring
 394 $\ell_i \phi_m \psi_a \geq \delta_k(t)$ makes each mode carry its share of the shortfall, so the test tightens as the gap grows. The
 395 allocation is then determined by equation (31):

$$(m, a)_i^* = \arg \max_{(m,a) \in \mathcal{A}(i,t)} \widetilde{SCV}_{i,m,a}, \tag{31}$$

396 over the admissible set $\mathcal{A}(i, t)$, with the Zone Z4 fallback when $\mathcal{A}(i, t) = \emptyset$. After execution, outcomes are
 397 folded back into policy through equation (32), the closed governance loop of Section 5.1:

$$\text{Policy}_{t+1} = f(\text{Policy}_t, \text{Performance}_t, \text{Incidents}_t, \text{CAD}_t, \text{Distribution}_t, \text{Feedback}_t), \tag{32}$$

398 Algorithm 1 states the per-task procedure, including the roster construction and the two intermediate policy
 399 statuses.

Algorithm 1 Online policy-aware allocation over staffed modes

Require: Task T_i ; time t ; ledger $L(t)$; policy twin $\mathcal{P}$; budgets $B(t)$; thresholds ρ ; duals $\lambda, \mu, \nu, \varrho, v$ from the last solution of (26)

Ensure: Selected staffed mode $(m, a)_i^*$

```

1: Encode  $T_i$  via (8);  $\ell_i \leftarrow \text{learn}_i d_i$ ;  $s \leftarrow \sigma(i)$ ;  $k \leftarrow \kappa(i)$ ;  $S_i^{\min} \leftarrow \text{risk}_i \text{crit}_i$ 
2:  $\mathcal{A} \leftarrow \emptyset$ ;  $\delta_k \leftarrow (30)$ 
3: for each  $m \in \mathcal{M}$  do
4:    $z \leftarrow \mathcal{P}(T_i, m, \cdot, t)$  evaluated on the mode-level rules ▷ join of (13)
5:   if  $z = \text{prohibit}$  then
6:     continue
7:   end if
8:    $\mathcal{A}_{i,m} \leftarrow$  feasible rosters for  $m$  from available  $w \in \mathcal{W}_k$ , with  $\psi_a$  by (4),  $\pi_{a,g}^{\text{dev}}$  by (5)
9:   if  $z = \text{restrict}$  then
10:    prune  $\mathcal{A}_{i,m}$  to rosters satisfying the firing rules' roster guards
11:   end if
12:   if  $z = \text{allow-with-oversight}$  then
13:    append the named reviewer to each roster; add reviewer time to  $\text{Cost}$ ; raise  $\phi_m$  by the assessed retained-judgment share
14:   end if
15:   for each  $a \in \mathcal{A}_{i,m}$  do
16:     Query the ten agents for  $Q, S, P, Eq^{\text{srv}}, Tr, \text{Cost}, En, Pr$  and  $D^{\text{skill}}, D^{\text{fall}}, D^{\text{acct}}, D^{\text{dep}}, D^{\text{trans}}$ 
17:      $CAD_{i,m,a} \leftarrow (6)$ ;  $SCV_{i,m,a} \leftarrow (25)$ ;  $\Delta_s^{\text{res}}(m, a) \leftarrow (28)$ 
18:     if  $(m, a)$  satisfies (29) then
19:        $\widetilde{SCV}_{i,m,a} \leftarrow (27)$ ;  $\mathcal{A} \leftarrow \mathcal{A} \cup \{(m, a)\}$ 
20:     end if
21:   end for
22: end for
23: if  $\mathcal{A} = \emptyset$  then
24:    $(m, a)_i^* \leftarrow$  Zone Z4 human-reserved roster; record the empty-set event; update  $L(t)$ ; return
25: end if
26:  $(m, a)_i^* \leftarrow \arg \max_{(m,a) \in \mathcal{A}} \widetilde{SCV}_{i,m,a}$  ▷ Equation (31)
27: Report  $w_9^*(i)$  by Proposition 3 to the allocation record
28: Execute; observe outcomes;  $\Lambda_k \leftarrow \Lambda_k + \ell_i \phi_m^* \psi_a^*$ 
29: For each  $w \in a^*$ :  $c_w \leftarrow c_w + \ell_i \phi_m^* \omega_{a^*,w}$ ; re-evaluate stage  $k(w, t)$  by (2)
30: Update  $L(t+1)$ , accumulated CAD,  $\Pi_{k,g}$ ,  $\Delta_g$ ,  $U_r^{\text{used}}$ ; emit to the feedback bus via (32)

```

400 Per task, Algorithm 1 performs at most seven policy queries and $\sum_m |\mathcal{A}_{i,m}|$ agent estimates, so the
 401 online decision is linear in arrivals and the periodic program is the bottleneck. The rule is exact only when
 402 the duals are exact and the state has not drifted (Section 7.3).

403 4.6 Feasibility of the Capability Budget and the Intake Requirement

404 A capability budget the task stream cannot supply is not demanding but infeasible. Proofs of the four
 405 propositions are in Appendix S8.

406 PROPOSITION 1 (Feasibility of the capability budget). Let $\Phi_k(\Delta T) = \sum_{i \in \mathcal{T}_k(\Delta T)} \ell_i$ be the domain's
 407 total developmental content in the period, and let

$$\bar{\zeta}_k = \max \{ \phi_m \psi_a : (m, a) \text{ admissible for some } i \in \mathcal{T}_k \}$$

408 be the largest developmental credit any admissible staffed mode delivers. Then equation (16) is satisfiable
409 only if the condition in equation (33) holds:

$$B_k(t) \leq \Phi_k(\Delta T) \bar{\zeta}_k. \quad (33)$$

410 Where the estimator (17) is attained by its second argument, this is equivalent to a ceiling on attrition
411 given by equation (34):

$$r_k(t) \leq r_k^{crit} = \frac{\Phi_k(\Delta T) \bar{\zeta}_k}{\eta_k N_k(t) h_k}, \quad (34)$$

412 and symmetrically to ceilings on η_k and h_k and a floor on the task volume.

413 Condition (33) binds harder than it appears: machine assistance and supervision ratios both reduce $\bar{\zeta}_k$,
414 and safety floors remove the high- ϕ_m candidates on precisely the hazardous tasks the budget most wants.
415 When it fails, the framework raises an infeasibility flag, clamps B_k to $\Phi_k \bar{\zeta}_k$, publishes the shortfall as a
416 capability deficit, and reports the four restoring actions: relax the supervision ratio where risk permits,
417 procure less autonomous tooling, expand the supervised task stream, or lower η_k as a recorded decision to
418 let the domain shrink. The framework also needs a hiring lever, which the next proposition supplies.

419 PROPOSITION 2 (Capability intake requirement). Let Θ_k be the clock hours a developing practitioner
420 works on domain tasks per budget period, $\bar{\phi}_k$ the mean human developmental share of the staffed modes
421 the domain selects, and $\bar{\omega}_k$ the mean work share a developing practitioner holds in those rosters. Then the
422 expected time for one developing practitioner to reach competence is given by equation (35):

$$\tau_k = \frac{h_k^{raw}}{\Theta_k} \cdot \frac{1}{\bar{\phi}_k \bar{\omega}_k}, \quad (35)$$

423 and in steady state the domain must carry the intake cohort defined in equation (36):

$$\hat{n}_k = \eta_k N_k(t) r_k(t) \tau_k \quad (36)$$

424 developing practitioners for (16) to be met.

425 Equation (35) factorizes formation time into the solo-practice time h_k^{raw}/Θ_k that certification assumes
426 and a dilution factor $1/(\bar{\phi}_k \bar{\omega}_k)$ imposed by machine assistance and shared supervision. A city cannot see
427 the dilution factor without this framework; Section 6.6 evaluates it at 3.38, turning a 1.2-year formation
428 into a 4.05-year one.

429 4.7 What the Price Does and What the Constraint Does

430 A framework named after a priced construct owes an account of how much of its behavior the price
431 produces.

432 PROPOSITION 3 (Price and constraint separation). For task i with admissible set $\mathcal{A}(i, t)$, let $\Xi(w_9)$
433 denote the arg max of (25) with the debt weight set to w_9 and $w_1, \dots, w_8$ held in fixed ratio, which is
434 without loss of generality because the arg max is invariant to positive rescaling of the objective. Define the
435 threshold $w_9^*(i)$ in equation (37):

$$w_9^*(i) = \inf \{ w_9 \geq 0 : \Xi(w_9) \neq \Xi(0) \}, \quad (37)$$

436 with $w_9^* = \infty$ when no finite weight changes the selection. Then for every $w_9 < w_9^*(i)$ the selected staffed
437 mode is the one an objective that ignores Civic Automation Debt entirely would select, and any difference
438 in the realized allocation is attributable to the constraints of (26) rather than to the price.

439 We report w_9^* with every allocation. A domain with $w_9 \ll w_9^*$ is the division of labor the framework
440 intends: the objective describes the liability while a hard constraint does the work. Section 6.2 computes
441 $w_9^* = 0.313$ for the worked case against a default of 0.22.

PROPOSITION 4 (Between-rebalance violation bound). Suppose the online rule of (31) runs on a window of n arrivals with the admissibility test (29) evaluated on the post-execution state, and let $\bar{\delta}_g = \max_{i,m,a} \delta_g^{disp}(i, m, a)$ and $\bar{u}_r = \max_{i,m,a} u_{i,m,a,r}$. Then at the end of the window the bounds in equation (38) hold:

$$\Delta_g \leq \tau_g + \bar{\delta}_g \quad \forall g, \quad \sum_i \sum_{m,a} x_{i,m,a} u_{i,m,a,r} \leq U_r \quad \forall r, \quad (38)$$

independently of n , and the resource bound is exact.

Proposition 4 lets the just-transition and resource constraints be enforced online. One task of overshoot is a weak guarantee; Section 7.3 states what a stronger one requires.

4.8 Adaptive Oversight Zones

Task classes are assigned to the four oversight zones of Table 3, which also fixes the staffed modes each zone admits.

Table 3. Adaptive Oversight Zones and the staffed modes each admits. Zone membership is evidence-conditioned by the predicates of Table 4, not a fixed sector label. ϕ^{Z3} defaults to 0.5.

Zone	Allocation principle	Admitted staffed modes
Z1 Autonomous	High-confidence, low-rights-impact tasks may be performed autonomously.	All $m \in \mathcal{M}$; any feasible roster including the null roster.
Z2 Augmented	AI or robots perform substantial work while a human remains meaningfully involved.	m with $\phi_m > 0$; any feasible roster.
Z3 Human Authority	AI may advise, but a designated human authority makes or validates the decision.	m with $\phi_m \geq \phi^{Z3}$; roster must name a competent decision-maker of record who may depart from the recommendation.
Z4 Human Reserved	Automation is restricted for reasons of dignity, democratic authority, capability preservation, or unacceptable systemic risk.	$m \in \{H, H+R\}$ with the robot as a tool under direct human control; roster must contain a competent practitioner.

The tiering, which formalizes the human-in-, on-, and out-of-the-loop taxonomy echoed in the EU AI Act (European Parliament and Council of the European Union, 2024), is old; what is new is its coupling to evidence through the predicates of Table 4. Promotion requires every condition *and* an affirmative panel decision, while demotion fires on any single condition, because evidence that automation works should not expand it without a political act, while evidence that it fails should contract it at once.

A zone change is recorded as a versioned policy update, and restoration after an automatic demotion always requires a panel decision. Appendix S8.4 collects the notation.

5 GOVERNANCE, CONTESTABILITY, LAWFULNESS, AND DATA PROTECTION

5.1 Policy Feedback and Weight Legitimacy

Governance is a closed loop: outcomes are evaluated by group, following (32), and feedback modifies policy parameters through authorized processes. The system operationalizes values set through legitimate

Table 4. Zone transition predicates. W is a rolling window, four budget quarters by default. Promotion requires all promotion conditions plus an affirmative panel decision; demotion fires on any single demotion condition and is recorded as a versioned policy update. Thresholds are panel-set; the values shown are the defaults listed in Appendix S2 (Table S3).

	Promotion (one zone toward Z1), all required	Demotion (one zone toward Z4), any sufficient
1	No incident of severity 3 or above in W , and incident rate below the class baseline.	One incident of severity 3 or above, or incident rate above $1.5 \times$ the class baseline.
2	Upheld-appeal rate ≤ 0.02 over W .	Upheld-appeal rate > 0.05 over W (the systemic trigger of Section 5.2).
3	Accumulated CAD for the class non-increasing over W .	Accumulated CAD for the class rising for two consecutive quarters.
4	$Res_s^{(n_s)} \geq 1.15 \rho_s$ throughout W .	$Res_s^{(n_s)} < \rho_s$ at any evaluation.
5	$\Lambda_k \geq B_k$ in the last closed budget period.	$\Lambda_k < \bar{B}_k(t)$ for two consecutive quarters, or the infeasibility flag of Proposition 1 is raised.
6	Affirmative Weight Review Panel decision.	Input-distribution drift on the class exceeding a population stability index of 0.25, or a new <i>restrict</i> or <i>prohibit</i> rule entering the twin.

governance rather than deciding them, and fails visibly when those processes have not spoken, leaving an unset $\theta_{k,g}$ or τ_g flagged rather than quietly defaulted.

The weights, budgets, access targets, group coarsening, transition ceilings, zone thresholds, and contingency parameters are set by a standing *Weight Review Panel* meeting quarterly and after any stress event. It combines a deliberative process for the rights- and distribution-facing parameters with technical review for the operational ones, following co-creation designs shown to raise trust in municipal AI (Phothong et al., 2026; Nechesov and Ruponen, 2024), and records every decision as a versioned update to the Policy Digital Twin, so any allocation traces to the configuration that authorized it. Organized labor must sit on it, since it has a direct stake in ϕ_m , $B_k(t)$, $\theta_{k,g}$, and τ_g (Callari et al., 2024; International Labour Organization, 2015); so must workers outside the payroll (De Stefano, 2016; Wood et al., 2019), whom $N_k(t)$ does not see and no incumbent representative has an incentive to raise; and so must a legal office that keeps the rule encoding current as law changes.

The panel's authority has a limit that no choice of membership removes. Because (25) is a weighted sum, every allocation the panel can reach by setting weights lies on the convex hull of the attainable set, and allocations in non-convex regions of the frontier, which may include the compromise a deliberative process would actually choose, are unreachable whatever it decides (Osika et al., 2023). Its authority is therefore bounded by the aggregation function rather than by its mandate, which is a constraint on the legitimacy claim rather than a technical footnote. Chebyshev or ε -constraint formulations would remove the bound at some cost in interpretability, and the hard constraints of (26) already act as ε -constraints for the constructs a city judges non-negotiable.

Liability follows the zones. In Z3 and Z4 the human decision-maker of record forms a traceable chain; in Z1 harms trace to the unit that certified the automated component; and in Z2 liability is apportioned by the evidentiary record of which decision steps each party controlled, since assigning Z2 harms wholly to the vendor would contradict the human-oversight logic of the EU AI Act (European Parliament and Council of the European Union, 2024).

5.2 The Contestability Procedure

Contestability in public-sector algorithms must be a first-class design requirement rather than an audit afterthought (Moon and Guha, 2026; Alfrink et al., 2023), and a system reporting an appeal-resolution rate without specifying an appeal procedure is measuring something it has not built. We specify four elements; Appendix S1 states each in full, and Table S1 fixes the time limits. *Standing* extends to affected residents and to any worker whose allocation, developmental access, or displacement is at issue, independently

of contract status, with the municipal ombudsman able to act on its own motion against a pattern — the systemic-challenge route missing from most deployed systems (Vaccaro et al., 2019). *Windows* are twenty working days to lodge and twenty to resolve, shorter with the outcome suspended where a decision is rights-affecting, and longer for worker appeals on access or displacement, which may surface only when a budget period closes. An *evidentiary record* written by every allocation holds the roster set before and after policy filtering, each candidate's term vector and *CAD* decomposition, the binding constraint, the duals in force, w_g^* , the configuration version, and the accountable human, and is disclosable to an appellant with standing. This is technological due process (Citron, 2008) made operational, and recording the configuration version is what makes an appeal against a *policy* possible (Almada, 2019). The *remedy* is re-decision under Zone Z3 by a designated human who may depart from the recommendation on the record, coupled to a systemic trigger: an upheld-appeal rate above 0.05 over the rolling window demotes the class one zone by predicate 2 of Table 4 and refers it to the panel.

The evidentiary record and appeals procedure are separable from the optimizer, and are the framework's most immediately deployable component: a municipality could run them on its existing allocation practice and gain most of the accountability benefit before any optimization is built.

5.3 Lawfulness of the Capability Access Constraint

A mechanism that allocates scarce developmental work against a composition target defined over gender or district is making a decision that equality law regulates. We give the analysis for EU law, which is the most developed.

Directive 2000/78/EC (Council of the European Union, 2000) and Directive 2006/54/EC (European Parliament and Council of the European Union, 2006) prohibit direct discrimination in access to employment, vocational training, and working conditions, and both contain a positive-action provision permitting measures to prevent or compensate for disadvantages linked to a protected ground. The Court of Justice has drawn the boundary in a line of cases beginning with *Kalanke* (C-450/93), which held unlawful a rule giving automatic and unconditional priority to women where candidates were equally qualified, and *Marschall* (C-409/95), which upheld a materially similar rule that carried a saving clause requiring individual assessment. *Abrahamsson* (C-407/98) confirmed that preference for a candidate who is merely sufficiently rather than equally qualified exceeds the exemption. *Badeck* (C-158/97) is the closest authority to our case, because it concerned the reservation of *training places* rather than posts, and the Court accepted such a measure where it did not deprive the other group of the opportunity to train elsewhere. Four design features of the Capability Access Constraint are chosen against that boundary rather than assumed past it.

1. *The constraint binds an aggregate, not an assignment.* Equation (20) constrains the share of a domain's protected practice hours over a full budget period. No individual task is assigned to a practitioner because of a protected characteristic, and no practitioner is excluded from a task by the constraint.
2. *The online channel is a tie-break, not a priority rule.* Between rebalances the target reaches the decision only through the dual $\nu_{k,g}$ in (27), which adds a term to a score that operational terms can outvote. Where two rosters differ materially on merit the merit term dominates, which is the *Marschall* saving clause expressed numerically rather than in prose.
3. *The constraint is elastic.* The slack $\varsigma_{k,g}$ in (26f) means an unattainable target produces a published shortfall, not a forced assignment, so a city can never be placed where the only feasible solution requires an unlawful allocation.
4. *Targets on non-protected grounds carry no such exposure.* The framework's most consequential distributional concern, the exclusion of contracted and platform-mediated practitioners, is a target on *contract status*, which is not a protected ground under either directive; nor is career stage. A municipality unwilling to set a gender or district target can still set the ones that address the framework's own worst failure mode.

Two limits remain. Whether a given $\theta_{k,g}$ is proportionate is a fact-specific judgment on the evidence of under-representation, which a municipality's legal office must make and no framework can make for it. And outside the EU the analysis differs, as the next subsection shows for the Gulf. What the framework contributes is that the target, its provenance, its tolerance band, and its realized shortfall are versioned, published, and disclosable under Section 5.2, so the proportionality judgment can be made on evidence and

544 challenged on the record. A composition-blind budget makes the same distributional choice invisibly and
545 cannot be challenged at all.

546 **5.4 A Second Jurisdiction: Workforce-Composition Regimes in the Gulf**

547 Section 5.3 treats the access target as a positive-action measure justified against a prohibition. In the Gulf
548 Cooperation Council states, composition targets over one ground, nationality, are instead an instrument of
549 labor-market policy. We take Saudi Arabia as the instance, because its regime is the best studied.

550 Saudi civil-service appointment is confined to nationals, with non-Saudis employed temporarily where no
551 qualified national applicant is available (Kingdom of Saudi Arabia, 1977). Private establishments, including
552 the contractors through which municipalities procure much of their sanitation, maintenance, and inspection
553 labor, are graded by the Nitaqat program into bands by the national share of their workforce, against
554 thresholds that vary with economic activity and establishment size; establishments in the lowest band lose
555 access to new work permits and visa renewals. Studying that design, Peck (2017) finds that the quota raised
556 national employment at a substantial cost in firm exit and in reduced employment at surviving firms, borne
557 mainly by firms that had no national employees when it began.

558 Read against (20), three features transfer directly. The target $\theta_{k,g}$ is set by policy rather than by the
559 mechanism; it is published and audited; and shortfall is penalized in grades rather than prohibited, so
560 a firm below its band loses privileges rather than its license to operate, which is the elastic slack $\varsigma_{k,g}$
561 of (26f) enforced administratively. One feature does not transfer. The proportionality inquiry of Section 5.3,
562 including the *Marschall* saving clause, has no direct analogue, because the question a regulator asks of a
563 nationality target is whether it is met, not whether it may lawfully be set. Targets on gender, district, or
564 contract status would need their own analysis under Saudi law, which we do not attempt.

565 The substantive difference is what is counted. Nitaqat counts posts, so a compliant domain can staff every
566 developmental task with an incumbent expatriate lead and form no national successor, which is the gap
567 ψ_a closes in (16). With g ranging over nationality, (20) converts a headcount obligation into a formation
568 obligation: a city can show not only that nationals hold posts but that they receive the protected practice
569 that makes them competent. Where composition targets are already law, the access constraint is less a novel
570 positive-action decision than an existing obligation made computable and contestable, which strengthens
571 the case for deploying it.

572 The Just Transition Constraint meets the same structure from the other side. The ILO–ESCWA projections
573 describe Gulf economies that depend heavily on migrant labor while public-sector employment is
574 held largely by nationals, and name migrants among the groups most at risk of exclusion in an AI
575 transition (International Labour Organization and United Nations Economic and Social Commission for
576 Western Asia, 2026). The population (21) cannot see is therefore large in a Gulf municipality, predominantly
577 migrant, and employed through the contractors Nitaqat grades, so a ceiling τ_g on the payroll protects the
578 group least exposed. The regulator already computes each contractor's workforce composition, so part
579 of the substrate for extending (21) beyond the payroll exists; whether a municipality may use it for that
580 purpose is a question for the Personal Data Protection Law (Kingdom of Saudi Arabia, 2021), which plays
581 here the role the GDPR plays below.

582 **5.5 Data Protection and the Civic Capability Ledger**

583 The ledger of (10) holds per-practitioner practice totals indexed to worker groups defined over gender,
584 contract status, and district, retained for a decade and used to allocate work. Under the GDPR (European
585 Parliament and Council of the European Union, 2016) that is high-risk processing, so a data protection
586 impact assessment under Article 35 is mandatory before deployment. The design assumes four measures,
587 set out in Appendix S9: minimization by aggregation, since the allocator needs only the career-stage test
588 and aggregate shares; separation of protected attributes from the optimizer; small-cell suppression through
589 the $n^{\min}$ rule; and the Article 22 safeguards of the contestability procedure, whose Z3 re-decision is the
590 required human intervention. A city that places worker-affecting task classes in Z1 should therefore expect
591 to defend that under Article 22 and not only under the AI Act. None of this makes the ledger unproblematic.
592 It makes it a processing operation whose risks are documented, bounded, and challengeable, which is the
593 most a design can offer and less than a worker might reasonably want.

6 A WORKED ALLOCATION: MUNICIPAL BRIDGE INSPECTION

This section carries one allocation through the framework end to end. Every number in Sections 6.2 through 6.9 is exact deterministic arithmetic on the stated inputs and requires no simulation. The inputs are illustrative expert estimates rather than measurements; Section 6.8 maps where each conclusion survives.

6.1 The Task, the Domain, and the Candidate Set

T_i is one cycle of fatigue-crack assessment on a steel girder span, with profile $cog = 0.80$, $phy = 0.60$, $emp = 0.10$, $risk = 0.70$, $auth = 0.90$, $priv = 0.20$, $urg = 0.40$, $learn = 0.80$, $crit = 0.90$ and $d_i = 10$ hours, so by (9) its developmental content is $\ell_i = 0.80 \times 10 = 8$ qualified-practice hours. A structural determination of this kind carries a statutory sign-off requirement, so the Policy Digital Twin returns *prohibit* for every mode lacking a licensed human under the encoding of Section 3.4, leaving $\mathcal{M}' = \{H, H+A, H+R, H+A+R\}$.

The safety floor $S_i^{min} = risk_i crit_i = 0.70 \times 0.90 = 0.63$ removes unaided human inspection ($S = 0.55$) and human plus AI ($S = 0.60$), leaving $\{H+R, H+A+R\}$. Each surviving mode admits two rosters: a_0 , a single competent licensed inspector ($\psi_{a_0} = 0$), and a_1 , a competent lead with one developing inspector on an equal split ($\psi_{a_1} = 0.5$). Because $risk_i = 0.70$, the roster guard allows one developing practitioner per lead, capping ζ_k at $\phi_{H+R} \times 0.5 = 0.35$.

The domain is structural inspection of bridges and major structures: a portfolio on a routine cycle plus fracture-critical and in-depth inspections, generating $n_k = 2,100$ inspection events per year of mean duration 10 hours and mean learning value 0.80, so $\Phi_k = 2100 \times 8 = 16,800$ qualified-practice hours per year. The workforce is $N_k = 24$ competent practitioners with annual attrition $r_k = 0.12$.

6.2 Pricing the Staffed Modes

Table 5 gives the agents' normalized estimates for the eight staffed modes, the credited developmental share $\phi_m \psi_a$, and the five debt components; applying the default debt weights to (6) and the default objective weights to (25) gives the last two rows.

Read alone, Table 5 favors a fully hybrid team staffed by a lead alone: $H+A+R/a_0$ has the best quality, safety, and productivity, the lowest cost, and the highest *SCV* at 0.3426, despite the largest debt of the eight at 0.5004. Pricing debt does not by itself change this. With the other weights in fixed ratio, the arg max first changes at the threshold of Proposition 3,

$$w_9^* = 0.313, \quad (39)$$

where $H+A+R/a_1$ overtakes $H+A+R/a_0$, against a default $w_9 = 0.22$. The price narrows the gap between the two rosters from 0.0241 to 0.0072 objective units, closing 70% of it; on these inputs the reversal is produced by the hard constraint, not the price.

6.3 The Capability Constraint Binds

Certification at inspector grade requires $h_k^{raw} = 1,800$ supervised clock hours, so (18) gives $h_k = 0.80 \times 1800 = 1,440$; we target independent competence rather than statutory team-leader authority, which is a separate legal threshold, and note that h_k is the parameter to which the example is most sensitive. With $\eta_k = 1.2$ and $B_k^{min} = 2,000$ hours, Equation (17) yields

$$B_k = \max(2000, 1.2 \times 24 \times 0.12 \times 1440) = 4,976.64 \text{ qualified-practice hours per year.} \quad (40)$$

The budget requires a mean credited share of at least $4976.64/16800 = 0.2962$, and Proposition 1 holds with little room, at $\Phi_k \bar{\zeta}_k = 5,880$. The unconstrained allocation delivers nothing: every inspection sent to $H+A+R/a_0$ credits zero. A domain staffed entirely by its incumbents forms no successors.

6.4 Solving the Program and Reading Its Price

Restricted to this domain and its single binding constraint, program (26) has an optimum with at most two active staffed modes, because one coupling constraint plus the simplex constraint (26b) admits at most

Table 5. Worked bridge-inspection allocation. Normalized term valuations for the eight staffed modes, resulting Civic Automation Debt from Equation (6), and Sustainable Civic Value from Equation (25). Roster a_0 is a competent lead alone; roster a_1 adds one developing inspector at an equal work split. Shading marks the four staffed modes that survive the safety floor $S_i^{min} = 0.63$. Only D^{skill} depends on the roster; the other four debt components are mode-level.

Term	H		H + A		H + R		H + A + R	
	a_0	a_1	a_0	a_1	a_0	a_1	a_0	a_1
<i>Flow terms, measured on this execution window</i>								
Q service quality	0.72	0.69	0.86	0.83	0.80	0.77	0.91	0.88
S safety	0.55	0.53	0.60	0.58	0.88	0.86	0.90	0.88
P productivity	0.40	0.35	0.62	0.57	0.70	0.65	0.88	0.83
Eq^{srv} service equity	0.70	0.70	0.72	0.72	0.70	0.70	0.74	0.74
Tr trust	0.80	0.78	0.74	0.72	0.72	0.70	0.68	0.66
$Cost$ operating cost	0.85	0.95	0.62	0.72	0.58	0.68	0.50	0.60
En energy	0.20	0.20	0.28	0.28	0.55	0.55	0.60	0.60
Pr privacy risk	0.10	0.10	0.25	0.25	0.30	0.30	0.38	0.38
<i>Stock-change terms, settled later</i>								
ϕ_m human share	1.00	1.00	0.75	0.75	0.70	0.70	0.55	0.55
ψ_a developmental eligibility	0.00	0.50	0.00	0.50	0.00	0.50	0.00	0.50
$\phi_m\psi_a$ credited share	0.000	0.500	0.000	0.375	0.000	0.350	0.000	0.275
$D^{skill} = 1 - \phi_m\psi_a$	1.00	0.50	1.00	0.625	1.00	0.650	1.00	0.725
D^{fall} fallback erosion	0.00	0.00	0.15	0.15	0.25	0.25	0.35	0.35
D^{acct} accountability dilution	0.00	0.00	0.07	0.07	0.05	0.05	0.13	0.13
D^{dep} vendor dependency	0.00	0.00	0.30	0.30	0.35	0.35	0.50	0.50
D^{trans} transition burden	0.00	0.00	0.10	0.10	0.20	0.20	0.30	0.30
$CAD_{i,m,a}$ from (6)	0.2800	0.1400	0.3816	0.2766	0.4260	0.3280	0.5004	0.4234
$SCV_{i,m,a}$ from (25)	0.2324	0.2391	0.2783	0.2773	0.3108	0.3082	0.3426	0.3355
Survives $S_i^{min} = 0.63$?	no	no	no	no	yes	yes	yes	yes

635 two nonzero variables at an LP vertex. Table 6 enumerates every admissible pair reaching the required
 636 share.

Table 6. All two-mode mixes of admissible staffed modes reaching the required mean credited share $\phi_m\psi_a = 0.2962$. The enumeration is complete: no pair excluding roster a_1 can reach a positive credited share at all, and a single coupling constraint admits at most two active modes at an LP vertex. Bold marks the optimum.

Mix	Shares of tasks	Mean $\phi_m\psi_a$	Mean SCV
$H+R/a_1$ with $H+A+R/a_1$	28.30% / 71.70%	0.2962	0.3277
$H+R/a_1$ with $H+A+R/a_0$	84.64% / 15.36%	0.2962	0.3135
$H+R/a_1$ with $H+R/a_0$	84.64% / 15.36%	0.2962	0.3086

637 The optimum sends 28.30% of inspections to robot-assisted human teams and 71.70% to fully hybrid
 638 teams, and staffs a developing inspector on every one of them. The capability constraint's dual follows
 639 from the tie between the two active staffed modes, equation (41):

$$0.308240 + 8 \lambda_k(0.350) = 0.335452 + 8 \lambda_k(0.275) \implies \lambda_k = \frac{0.027212}{0.6} = 0.045353, \quad (41)$$

640 with the two SCV values carried at six decimals because their difference is the whole numerator. One
 641 additional qualified-practice hour therefore costs 0.0454 objective units at the margin. This is the marginal
 642 opportunity cost of tightening the constraint, not the value of an hour of practice, which the framework
 643 neither estimates nor needs; earlier presentations inverted the reading. Whether 0.0454 units per hour is
 644 worth paying is the panel's decision. Substituting into (27) reproduces the ranking without solving the

645 program, equation (42):

$$\begin{aligned}\widetilde{SCV}_{H+R/a_1} &= \widetilde{SCV}_{H+A+R/a_1} = 0.435229 \\ &> \widetilde{SCV}_{H+A+R/a_0} = 0.3426 \\ &> \widetilde{SCV}_{H+R/a_0} = 0.3108.\end{aligned}\quad (42)$$

646 The first equality holds by construction. The capability price reverses the winner: the same objective,
647 estimates, and weights now select a trainee-staffed team and move over a quarter of inspections to robot-
648 assisted human work, so the reversal falls first on *who is on the team* and only second on *which machines*
649 *are used*. Integrally, 594 tasks in $H+R/a_1$ deliver 4,976.40 hours and miss the budget by 0.24, while 595
650 deliver 4,977.00, so the integer optimum's mean SCV of 0.327742 sits 7.8×10^{-6} below the LP value of
651 0.327750.

652 6.5 What Preservation Costs and What It Buys

653 Unconstrained allocation yields a mean SCV of 0.3426 per task, or 719.5 objective units a year over
654 2,100 tasks; the constrained optimum yields 0.3277, or 688.3 units. The constraint costs 31.2 units a year, a
655 reduction of 4.34%, and over the 4,976.64 hours it purchases the average price is 0.0063 units per hour,
656 below the marginal 0.0454 as convexity requires. Table 7 decomposes which terms absorbed that cost,
657 because an aggregate figure hides whether a city bought its pipeline with money, with quality, or with
658 safety.

Table 7. Term-by-term decomposition of the cost of preservation in the worked domain: the unconstrained allocation (all tasks to $H+A+R/a_0$) against the constrained optimum. Cost, energy, and privacy risk are penalties in Equation (25), so a fall is an improvement for those three rows.

Term	Unconstrained	Constrained optimum	Change
Q service quality	0.9100	0.8489	−6.72%
S safety	0.9000	0.8743	−2.85%
P productivity	0.8800	0.7791	−11.47%
Eq^{srv} service equity	0.7400	0.7287	−1.53%
Tr trust	0.6800	0.6713	−1.28%
$Cost$ operating cost	0.5000	0.6226	+24.53%
En energy	0.6000	0.5858	−2.36%
Pr privacy risk	0.3800	0.3574	−5.96%
CAD civic automation debt	0.5004	0.3964	−20.78%
Credited share $\phi_m \psi_a$	0.0000	0.2962	from zero
SCV	0.3426	0.3277	−4.34%

659 The city pays mainly in operating cost, up 24.5% because every inspection carries a second person, and
660 productivity, down 11.5%, with smaller losses in quality and safety, both above the floor. Preservation
661 costing 4.34% of objective value and costing a quarter of the inspection budget are the same fact, but only
662 the second is the one a finance committee will contest.

663 What the city buys is a change of sign. Under the constraint the domain generates $4976.64/1440 = 3.456$
664 competent-equivalents a year against attrition of $24 \times 0.12 = 2.88$, a net gain of 0.576; without it, it forms
665 no one and loses 2.88 a year. The identity $B_k/h_k = \eta_k N_k r_k$ makes this replenishment rate independent of
666 h_k , even though feasibility is not.

667 6.6 Pipeline Arithmetic and the Intake Requirement

668 The optimum staffs a developing inspector on all 2,100 inspections, consuming 21,000 trainee clock
669 hours a year, each of which accrues only $4976.64/21000 = 0.237$ qualified-practice hours, because practice

is diluted by machine assistance ($\bar{\phi}_k = 0.5925$ at the optimum) and by sharing the human work with a lead ($\bar{\omega}_k = 0.5$). At $\Theta_k = 1,500$ domain clock hours per trainee-year, Proposition 2 gives the formation time in equation (43):

$$\tau_k = \frac{1800}{1500} \cdot \frac{1}{0.5925 \times 0.5} = 1.20 \times 3.376 = 4.05 \text{ years}, \quad (43)$$

and the intake requirement in equation (44):

$$\hat{n}_k = 1.2 \times 24 \times 0.12 \times 4.05 = 14.0 \text{ developing practitioners}. \quad (44)$$

Both routes agree: 21,000 trainee clock hours at 1,500 per trainee-year is exactly 14.0 posts, and Little's law at 3.456 completions a year and 4.05 years' residence gives the same, an identity that lets a city compute its intake from its task stream without modeling its pipeline.

The dilution factor is the finding, and Figure 3 draws it. Certification assumes 1,800 supervised hours, 1.2 years of solo practice; under the optimal machine-assisted supervised mix the same practitioner takes 4.05 years, 3.38 times longer, and a plan built on the certification figure would size intake at four trainees where fourteen are needed. No headcount plan surfaces this, because the dilution lives in $\bar{\phi}_k$ and $\bar{\omega}_k$, properties of the *allocation* rather than the workforce. A city that automates more aggressively forms each practitioner more slowly as well as forming fewer, and will not see the second effect until the first has happened.

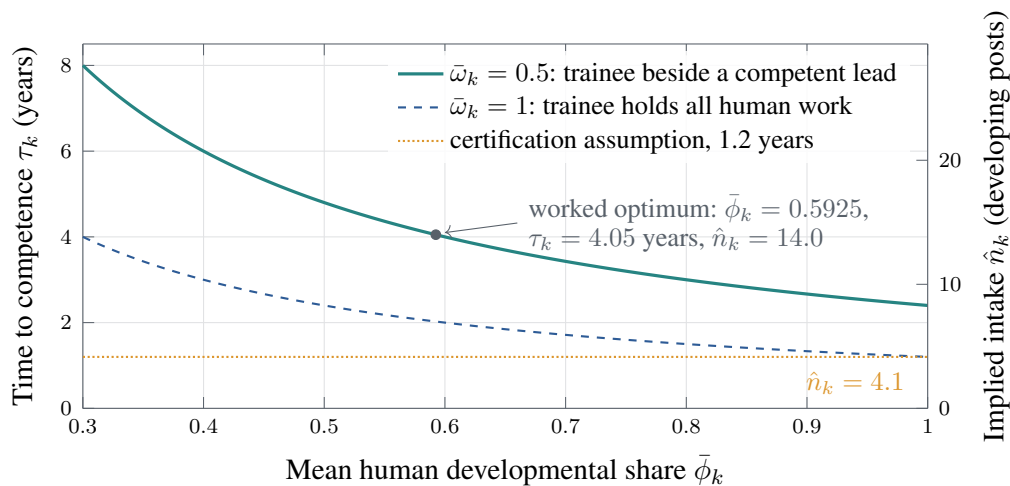


Figure 3. Time to competence under Equation (35) against the mean human developmental share of the selected staffed modes, at the worked domain's solo-practice time $h_k^{raw}/\Theta_k = 1800/1500 = 1.2$ years. The solid curve is a developing practitioner holding half of a roster's human work, as at the worked optimum of Section 6.4; the dashed curve is one holding all of it. The certification assumption of 1.2 years is reached only when $\bar{\phi}_k\bar{\omega}_k = 1$, that is, solo unassisted practice. The right axis is the steady-state intake of Equation (36), $\hat{n}_k = \eta_k N_k r_k \tau_k = 3.456 \tau_k$: the certification figure implies 4.1 developing posts and the worked optimum 14.0. Every point is Equation (35) evaluated at the article's stated values; nothing is measured.

The training-economics literature explains why no employer corrects this unprompted. Skills formed on the job are largely portable, so the employer that trains bears the cost while others can hire the result, and training is under-provided unless labor-market frictions let the trainer recoup it (Acemoglu and Pischke, 1998; Wolter and Ryan, 2011). The dilution factor raises the trainee time a city must fund per competent practitioner by the same 3.38, without appearing in any training budget.

6.7 Who Receives the Protected Work

The 4,976.64 protected hours must go to someone, and (16) does not say to whom. Suppose the 24 incumbents are 21 men and three women, 87.5% to 12.5%, while the qualified local labor pool stands at 68% to 32%. A composition-blind budget recruits through the channels that produced the incumbents

and distributes protected practice in proportion, sending $0.125 \times 4976.64 = 622$ hours a year to the under-represented group, or $622/1440 = 0.432$ competent-equivalents against a domain-wide 3.456. The cadre's composition is not merely preserved but re-founded, one replacement cycle at a time.

Constraint (20) intervenes on the distribution rather than the size. Setting $\theta_{k,g}$ to the labor-pool composition of 0.32 with $\varepsilon_k = 0.05$ requires $\Pi_{k,g} \geq 0.27$, or at least 1,344 hours a year, equal to 0.933 competent-equivalents: 2.16 times the composition-blind rate, achieved without changing B_k , the weights, or any estimate, and meaning 3.8 of the 14 developing posts of (44) rather than 1.8. The point is not the multiplier but that who wins is set by a published parameter a panel chose and can be asked to defend. Table 8 gives the whole distribution both ways; because protected practice is scarce at the optimum the access constraint binds with equality, so every cell is a determinate consequence of B_k and the panel's choice of $\theta_{k,g}$ and ε_k .

Table 8. Distribution of the $B_k = 4,976.64$ protected practice hours in structural inspection, derived by exact arithmetic. “Composition-blind” distributes protected practice in proportion to the incumbent composition; “Under (20)” applies the binding floor $\Pi_{k,g} \geq \theta_{k,g} - \varepsilon_k$ with $\theta_{k,g} = 0.32$ and $\varepsilon_k = 0.05$. Competent-equivalents are hours divided by $h_k = 1,440$. Groups are the marginal partition on gender and contract status, with the intersection suppressed under the n^{min} rule. The contracted row is not a result and not a modeling choice: its share is structurally zero for any $\theta_{k,g}$, because non-payroll practitioners are absent from N_k and ineligible as roster members under (19).

Group	Incumbent	$\theta_{k,g}$	Composition-blind			Under (20)		
			$\Pi_{k,g}$	h/yr	c.eq.	$\Pi_{k,g}$	h/yr	c.eq.
Men, permanent	0.875	0.680	0.875	4,354.56	3.024	0.730	3,632.95	2.523
Women, permanent	0.125	0.320	0.125	622.08	0.432	0.270	1,343.69	0.933
Contracted, any gender	—	unset	0.000	0.00	0.000	0.000	0.00	0.000
Total on payroll	1.000	—	1.000	4,976.64	3.456	1.000	4,976.64	3.456

One group is invisible in these numbers, and the third row of Table 8 is the only honest way to show it. If 40% of this domain's inspection hours were delivered by contracted rope-access technicians off the municipal payroll, they would not appear in N_k , would not be eligible roster members, and would receive a structurally zero share whatever $\theta_{k,g}$ is, with their displacement unrecorded. That is a property of the data substrate rather than of the allocation, so no setting of the access constraint reaches it; Sections 7.5 and 7.7 treat it as the framework's most consequential distributional limitation.

6.8 Sensitivity and the Feasibility Frontier

Every conclusion above rests on expert estimates, so Table 9 sweeps the six load-bearing parameters one at a time, reporting the optimum, the dual, the cost of preservation, and whether Proposition 1 holds, all by exact arithmetic.

Four things follow, and two of them are uncomfortable. First, the qualitative staffing result holds across the frontier and its magnitude does not. No lead-only roster delivers any credit and the budget is strictly positive, so some developmental staffing occurs at every feasible parameter value we swept; but across the eight feasible rows labeled “no reversal” the constrained optimum retains the lead-only roster on between 1.3% and 46.1% of tasks, and the dual falls from 0.0454 to between 0.0005 and 0.0033. A binding capability budget always places developing practitioners on some of the work, not on essentially all of it; the latter is a property of the baseline parameterization. Second, the *mode* reversal is not robust: it requires $\phi_{H+A+R} \leq 0.55$ and $\psi_{a_1} \leq 0.5$, and relaxing either lets the fully hybrid mode nearly satisfy the budget alone, collapsing the dual by an order of magnitude. That is correct behavior, since a mode that develops people should not be penalized, but the headline mode shift is a statement about one estimate of ϕ_{H+A+R} . Third, feasibility is fragile. The baseline sits at 85% of the critical value on all three axes Proposition 1 bounds, so raising h_k^{raw} to 2,400 hours, well within a stricter licensing regime, or attrition to 0.16, which a retirement wave produces easily, makes the budget unattainable: the domain has roughly 15% headroom on three separate axes. Fourth, the cost of preservation ranges from 0.33% to 9.60% across the swept region, and wherever this article quotes 4.34%, Table 9 is the range that should accompany it.

Table 9. One-at-a-time sensitivity of the worked allocation. Baseline row in bold. “Infeasible” means Proposition 1 fails, so the budget exceeds what the task stream can supply and the escalation of Section 4.6 is triggered. “No reversal” means the constrained optimum retains the unconstrained lead-only roster on some tasks, so the capability price does not change the staffing decision.

Parameter	Value	Optimum	λ_k	Cost	Status
h_k^{raw} (clock h)	1200	28.2% $H+A+R/a_0$ + 71.8% $H+A+R/a_1$	0.0033	1.50%	no reversal
	1500	10.2% $H+A+R/a_0$ + 89.8% $H+A+R/a_1$	0.0033	1.88%	no reversal
	1800	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	2127	boundary of Proposition 1	—	—	critical
	2400	—	—	—	infeasible
	3000	—	—	—	infeasible
r_k (per year)	0.06	46.1% $H+A+R/a_0$ + 53.9% $H+A+R/a_1$	0.0033	1.13%	no reversal
	0.09	19.2% $H+A+R/a_0$ + 80.8% $H+A+R/a_1$	0.0033	1.69%	no reversal
	0.12	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.14	94.1% $H+R/a_1$ + 5.9% $H+A+R/a_1$	0.0454	9.57%	near-critical
	0.1418	boundary of Proposition 1	—	—	critical
η_k	0.8	28.2% $H+A+R/a_0$ + 71.8% $H+A+R/a_1$	0.0033	1.50%	no reversal
	1.2	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	1.4	94.1% $H+R/a_1$ + 5.9% $H+A+R/a_1$	0.0454	9.57%	near-critical
	1.6	—	—	—	infeasible
ϕ_{H+R}	0.55	—	—	—	infeasible
	0.60	84.9% $H+R/a_1$ + 15.1% $H+A+R/a_1$	0.1515	9.60%	reversal
	0.70	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.85	14.2% $H+R/a_1$ + 85.8% $H+A+R/a_1$	0.0188	3.02%	reversal
ϕ_{H+A+R}	0.45	57.0% $H+R/a_1$ + 43.0% $H+A+R/a_1$	0.0241	7.00%	reversal
	0.55	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.60	1.3% $H+A+R/a_0$ + 98.7% $H+A+R/a_1$	0.0023	1.62%	no reversal
	0.70	15.4% $H+A+R/a_0$ + 84.6% $H+A+R/a_1$	0.0009	0.63%	no reversal
ψ_{a_1} (supervision)	0.40	—	—	—	infeasible
	0.4232	boundary of Proposition 1	—	—	critical
	0.50	28.3% $H+R/a_1$ + 71.7% $H+A+R/a_1$	0.0454	4.34%	baseline
	0.67	19.6% $H+A+R/a_0$ + 80.4% $H+A+R/a_1$	0.0005	0.33%	no reversal

6.9 Two Parameter Shocks: Stability and Infeasibility

Two displacements are worth carrying through, because they are the two regimes Proposition 1 separates and a city will meet both. Both are exact arithmetic on the model, not simulation output; Appendix S10 gives every step.

Shock one: a retirement wave inside the feasible region. Raising attrition from 0.12 to 0.14, inside the critical rate of 0.1418, expands the budget to 5,806.08 hours and the required credited share from 0.2962 to 0.3456. The same pair of modes still brackets the requirement, so λ_k stays at 0.0454 and only the mix moves: robot-assisted human work rises from 28.3% to 94.1%, the cost of preservation from 4.34% to 9.56%, and the domain operates at 98.7% of its ceiling. This is the stabilizing direction the capability feedback is designed to have. The arithmetic does *not* establish that an implementation updating duals at a finite rebalance cadence converges to this point rather than oscillating around it; no result in this article does, and Section 7.3 states what would.

Shock two: a retirement wave outside it. At attrition 0.25 the budget becomes 10,368 hours against the ceiling of 5,880, a deficit of 4,488 hours or 3.12 competent-equivalents a year that *no* allocation of this task stream can form. Of the restoration options of Section 4.6, permitting two developing practitioners per lead leaves a residual deficit of 2,528 hours; expanding the supervised stream to 3,703 inspections a year closes it; and lowering η_k to 0.68 closes it by deciding not to replace the full cadre, a structural contraction that should be recorded as one. The lesson concerns task structure rather than recruitment. The binding quantity is $\Phi_k \bar{\zeta}_k$, the practice the task stream can carry: a municipality facing this shock could recruit thirty trainees and still fail, because the stream supports at most 14 supervised trainee posts at the prevailing automation level.

7 DISCUSSION

7.1 An Analytic Model of Debt Accumulation

The debt components of (6) are per-task quantities; what a city experiences is their accumulation, and its shape distinguishes a strategy that replenishes a stock from one that does not. Let $\pi_j(t) \in [0, 1]$ be the fraction of component j 's required replenishment delivered at time t , let debt accrue in proportion to the unmet fraction, and let each component approach its own asymptote with its own time constant, equation (45):

$$\dot{C}_j(t) = c_j [1 - \pi_j(t)], \quad \pi_j(t) = \pi_{j,\infty} (1 - e^{-t/\tau_j}). \quad (45)$$

Integrating from $\pi_j(0) = 0$ and weighting by the debt weights of (6) gives the total debt trajectory in equation (46):

$$C(t) = \sum_j w_j c_j \left[(1 - \pi_{j,\infty}) t + \pi_{j,\infty} \tau_j (1 - e^{-t/\tau_j}) \right], \quad w_j \in \{\alpha, \beta, \gamma, \delta, \epsilon\}. \quad (46)$$

Modeling all five components rather than D^{skill} alone matters because different mechanisms drive each $\pi_{j,\infty}$: the hard constraint (16) drives skill formation to one, with $\tau_{skill} = 4.05$ years taken from Proposition 2; the reserve (24) drives fallback to one; revisable zones and (21) drive accountability and transition high; and *nothing in the framework drives vendor dependency*, which keeps accruing linearly. Table 10 fixes every parameter and Figure 4 plots (46) at exactly those values.

Table 10. Per-component replenishment parameters for Equation (46), one row per strategy, reported as $\pi_{j,\infty}$ with τ_j in years in parentheses. The accrual rate is $c_j = 10$ index units per year for every component of every strategy, except that Cost/Performance carries $c_{acct} = 14$ and $c_{dep} = 16$. These are the article's illustrative choices, not measured quantities; they are tabulated so that Figure 4 is reproducible by direct substitution rather than asserted. Only $\tau_{skill} = 4.05$ is derived, from Proposition 2. The final two columns are computed from the row, not fitted to the plot.

Strategy	$\pi_{skill,\infty}$ (τ)	$\pi_{fall,\infty}$ (τ)	$\pi_{acct,\infty}$ (τ)	$\pi_{dep,\infty}$ (τ)	$\pi_{trans,\infty}$ (τ)	Residual slope	$C'(10)$
Automation-First	0	0	0	0	0	10.00	100.00
Cost/Performance	0	0	0	0	0	11.44	114.40
Capability Matching	0.35 (4.05)	0.30 (2.0)	0	0	0	8.36	88.54
Ergonomics-Aware Role Allocation	0.40 (4.05)	0.45 (2.0)	0	0	0	7.89	85.02
Human-First	0.15 (4.05)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	1.00 (0.5)	2.38	28.96
CivicWorkOS	1.00 (4.05)	1.00 (1.0)	0.85 (1.5)	0.10 (2.0)	0.95 (1.0)	1.45	31.51

Three things follow from Figure 4, and one of them is unfavorable to the framework.

No strategy here bounds debt, CivicWorkOS included. Its saturating components contribute a bounded 17.98 index units, 11.34 of them from skill formation, but the residual accrues at 1.45 units a year without limit, 1.08 of it vendor dependency. A capability budget converts *skill* debt from unbounded to bounded and says nothing about the other components except through outvotable weights. Vendor dependency is a procurement variable the allocation mechanism does not control; it belongs to the governance of procurement, where the specification of an algorithmic system is itself a policy decision requiring administrative process (Mulligan and Bamberger, 2019), and pretending otherwise would repeat the category error Section 3.6 corrects for reskilling coverage.

Human-First beats CivicWorkOS on this metric for thirteen years. Keeping humans on everything replenishes fallback, accountability, dependency, and transition at once, while CivicWorkOS fills its pipeline over a 4.05-year time constant. The curves cross at $t = 13.23$, both at 36.73 index units, and by year twenty stand at 46.89 against 52.89, because Human-First never closes its skill gap: maximizing human share is not forming practitioners. On a ten-year horizon and this index alone a city should prefer Human-First, and would pay for it in operational terms the index omits, which is why debt is one construct among thirteen rather than the objective.

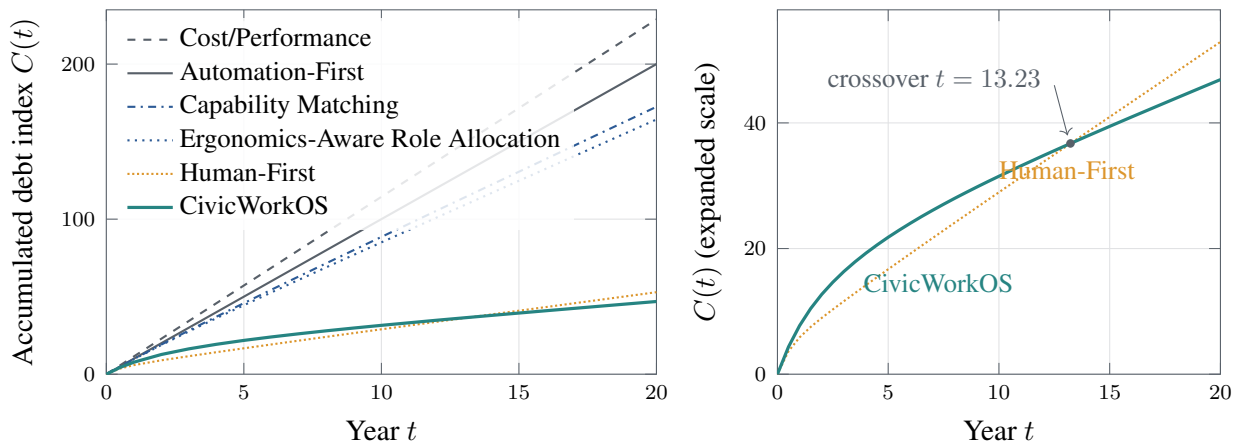


Figure 4. Accumulated Civic Automation Debt under Equation (46) at the per-component parameters of Table 10. Every plotted point is that equation evaluated at those parameters; nothing here is measured or simulated. *Left:* all six strategies over twenty years. Automation-First is normalized to 100 at year ten by setting $c_j = 10$ for every component, and Cost/Performance exceeds it because it carries elevated accountability and dependency accrual. *Right:* the two capability-forming strategies at an expanded scale. CivicWorkOS drives skill formation and fallback to full replenishment through hard constraints, with $\tau_{skill} = 4.05$ years derived from Proposition 2, but has no mechanism for vendor dependency, so its curve does not saturate. Human-First accumulates less debt than CivicWorkOS until year 13.23 and more thereafter, because its residual slope of 2.38 index units per year, almost all of it unformed skill, exceeds CivicWorkOS's 1.45, three-quarters of which is vendor dependency.

778 *The ordering of the curves is not evidence.* It follows analytically from the mechanisms each strategy
 779 enforces, so Figure 4 describes the debt accounting, not cities. Whether $\pi_{skill,\infty}$ can reach one across
 780 heterogeneous task streams without excessive Zone Z4 fallback is what the protocol of Section 7.4 would
 781 answer.

7.2 What the Framework Establishes, and What It Does Not

783 Section 6 establishes by arithmetic that the constraint system is computable on a realistic domain, that its
 784 constraints bind, that the price of the binding constraint is legible, and that the distributional question has a
 785 determinate answer with (20) and none without it; Section 6.8 bounds those results. None of that establishes
 786 that any input estimate is correct: the estimates are the authors' structured judgment, the elicitation protocol
 787 of Appendix S5 has not been run, and sensitivity analysis is a weaker substitute for measurement.

788 Nor is a property of the construction a finding. Because $-w_9CAD$ sits in the same weighted sum as P
 789 and $-Cost$ under hard constraints, any SCV -maximizing strategy sacrifices some productivity and cost
 790 for capability; that is a tautology, and it applies to hypotheses P2, P3, P5, and P9 too. The substance lies in
 791 magnitudes — whether the sacrifice is 4% or 40%, and whether the system is satisfiable at municipal scale
 792 — of which Section 6.8 bounds the first between 0.33% and 9.60% for one domain.

7.3 What a Guarantee for the Online Rule Would Require

794 Proposition 4 says nothing about how the online rule performs against the program it approximates.
 795 The bound to seek is the standard pair from online allocation: regret of order $O(\sqrt{T})$ in the number
 796 of arrivals T against the offline optimum of (26) over the period, together with a cumulative violation
 797 of (26e) and (26f) of the same order. Close relatives have both. When arrivals are in random order and
 798 each request is small relative to the budget, a primal–dual rule that prices a packing budget is $(1 - \epsilon)$ -
 799 competitive (Devanur and Hayes, 2009); dual mirror descent attains $O(\sqrt{T})$ regret without exceeding any
 800 resource under stochastic arrivals and a fixed competitive ratio under adversarial ones (Balseiro et al.,
 801 2023); the adversarial benchmark is the $1 - 1/e$ ratio of online budgeted matching (Mehta et al., 2007);
 802 and for constrained MDPs the duality theory (Altman, 1999) and a zero duality gap (Paternain et al.,

2019) license the same primal–dual construction. The small-request condition holds comfortably here: one inspection carries at most $8 \times 0.35 = 2.8$ credited hours against a budget of 4,976.64, a ratio of 5.6×10^{-4} .

Four features of (27) keep these results from applying directly. First, the duals are refreshed at the rebalance cadence rather than after each arrival, so the rule runs on stale prices and any bound must carry a tracking term. Second, the capability budget is a *covering* constraint that must be met by the end of the period, whereas those results bound *packing* constraints that must not be exceeded: a packing budget is protected by stopping, a covering one only by reserving future capacity, and the catch-up threshold (30) is a heuristic for that reservation, not an analyzed policy. Third, the constraint coefficients are endogenous, because ψ_a depends on career stage and assignments move practitioners across the threshold of (2), so the problem is non-stationary in exactly the state the constraint measures. Fourth, the access constraint is a ratio, and the staffed-mode action space is combinatorial in roster size. The second and third obstacles are specific to capability formation, and are where we expect the difficulty to lie.

7.4 A Specified Evaluation Protocol

The protocol described in this subsection is specified in full and *has not been executed*. It fixes the required evaluation in advance, so that it can be audited against a plan rather than one assembled after outcomes are known, and nothing in this article reports an outcome of it; Appendix S2 gives the complete specification. It is a discrete-event testbed over six municipal sectors and 803,100 tasks a year, run for ten years so that retirement waves and pipeline decay fall inside the horizon, comparing Automation-First, Cost/Performance, Human-First, and CivicWorkOS — all configurations of (25) — with the external baselines of Ranz et al. (2017) and Merlo et al. (2023), under paired Wilcoxon tests with Holm–Bonferroni control over sixty tests and seven stress scenarios, two of which Section 6.9 works out by exact arithmetic.

An evaluation in which four of six comparators are configurations of the proposed objective can show that the mechanism does what it was designed to do; it cannot show that it outperforms an independently designed alternative. The two external baselines are specified for that reason, and the comparison is incomplete until they are run; neither is implemented at the time of writing. The calibration portals also belong to high-income cities with few non-payroll workers, so the evaluation would understate the population the framework fails; an earlier version of this article wrongly called that bias conservative.

The hypotheses are *pre-specified*, not pre-registered, since a manuscript is not a timestamped registry entry; they will be lodged on the Open Science Framework before execution and the identifier cited in whichever article reports the outcome. Their thresholds were also set with the worked example's arithmetic in hand, a weaker form of pre-specification than setting them blind, and we claim no more.

7.5 Who Wins and Who Loses

The distributional question has four answers in this framework, and they are not equally comfortable.

The clearest winners are residents of a city whose critical services retain human fallback: constraint (24) makes an AI-service outage a degradation rather than an interruption, although a reserve sized for an outage is not one sized for a surge.

The second group is workers who receive protected developmental work, provided $\theta_{k,g}$ is deliberately calibrated rather than inherited. A capability budget rations a scarce good, and under a composition-blind budget the losers are precisely the under-represented groups, at 622 hours a year against a possible 1,344 in Section 6.7.

The third group is the one the framework serves worst. Workers outside the municipal payroll — contracted, agency, informal, or platform-mediated (De Stefano, 2016; Wood et al., 2019) — are absent from $N_k(t)$, ineligible as roster members, and unprotected by (21). Where they perform much of a city's sanitation, inspection, and logistics labor, CivicWorkOS as specified would preserve the permanent workforce while the contracted one absorbed the displacement, and report a satisfied access constraint while doing so; Table 8 prints the structurally zero row rather than hiding it, and Section 5.4 explains why that row is large in a Gulf city. No parameter setting fixes this, because it follows from a data substrate that presumes formal employment.

The fourth answer concerns automation itself, which CivicWorkOS legitimates as much as it constrains: a documented *SCV* computation, a satisfied budget, and a versioned panel decision can justify displacing

workers once the price is paid. Whether that is a feature or a hazard depends on who sets the weights, which is why panel composition is a design parameter and why Proposition 3 lets a city see whether its objective or its constraints are deciding. The framework is not neutral between contested outcomes; it makes a political choice explicit and traceable, and should be judged as such.

7.6 Threats to Validity

Construct validity. Qualified-practice hours are a proxy, and hours are not learning; without the retention functions of the skill-decay literature (Arthur et al., 1998; Tatel and Ackerman, 2025), the ledger overstates competence under intermittent practice. *Internal validity.* Most terms are reported by agents with an interest in their value, and ϕ_m , which enters four channels, comes from the agent whose interest is a larger budget; cross-validation against observed outcomes is only a partial remedy. *External validity.* One city scale and a high-income calibration base say nothing about a majority-informal workforce, where the problem is most severe.

7.7 Limitations

- *No measured inputs.* The estimates and parameters are the authors' own; the elicitation protocol that would replace them (Appendix S5) has not been run.
- *Weight legitimacy.* The quarterly panel of Section 5.1 requires deliberative processes a municipality would have to build, and some parameters move faster than any panel meets.
- *Linear scalarization.* The panel can reach only the convex hull of the attainable set (Section 5.1).
- *No guarantee for the online rule.* Beyond Proposition 4 there is no regret or cumulative-violation bound; Section 7.3 states the bound to seek and four obstacles to it, and safe reinforcement learning (García and Fernández, 2015) offers a further route.
- *Unmeasured tractability.* Program (26) is NP-hard in general, and no claim here depends on where it stops being tractable.
- *No envy guarantee.* The access constraint is a share constraint, not a fair-division mechanism (Bouveret et al., 2016), so members of one group can receive very different practice.
- *Privacy and gaming.* The measures of Section 5.5 bound but do not remove the ledger's exposure, and ϕ_m is the most gameable quantity in the framework.
- *Machine-checkability and jurisdiction.* Decidable predicates cannot capture the discretionary judgment that constitutes much of a public service (Lipsky, 2010), and Sections 5.3 and 5.4 cover two jurisdictions without amounting to legal advice in either.
- *Formal-employment presumption.* The framework preserves the visible payroll workforce and cannot see the rest. This is the limitation we would fix first.

8 CONCLUSIONS

The AI era requires a richer question than whether humans, agents, or robots work better, because the allocation itself changes what a city will be able to do, and who will be able to do it. CivicWorkOS treats urban automation as a governed portfolio rather than a sequence of independent efficiency decisions.

Its central proposition is that human expertise, fallback competence, accountability, and democratic control are renewable but depletable civic resources, and that automation consuming them without replenishment creates Civic Automation Debt. The concern has established lineages (Bainbridge, 1983; Cunningham, 1992; Burke et al., 2004); the contribution is a decision variable that names who performs the work, a capability constraint that accrues only to the people it is sized to form, an access constraint that makes the resulting distribution an explicit and contestable choice, and the formal results that say when the system is satisfiable and what to do when it is not.

The worked allocation shows what this buys in an end-to-end municipal case. A capability price of 0.0454 objective units per practice hour reverses the winning staffed mode, costs 4.34% of annual objective value, and turns a domain that would have formed nobody into one that generates 3.46 competent-equivalents a year against attrition of 2.88. An access constraint more than doubles the share of that formation reaching an under-represented group, from 622 to 1,344 hours a year, without changing the budget. And the

framework's own arithmetic produces a result no headcount plan surfaces: machine-assisted supervised practice stretches time to competence by a factor of 3.38, from 1.2 years to 4.05, so a city automating its junior work does not merely form fewer practitioners, it forms each remaining one more slowly, and a plan sized on certification hours provisions four trainee posts where fourteen are required (Figure 3).

Three limits bear restating. The framework bounds skill debt but not vendor-dependency debt, so Figure 4 does not saturate. Its feasibility margin in the worked domain is roughly 15% on three axes, so a stricter certification regime or an older workforce reaches the boundary rather than the optimum. And it cannot see workers who are not on a payroll, the population the exposure literature identifies as most exposed.

Future work should execute the elicitation protocol so the inputs are measured; prove regret and violation bounds for the online rule (Section 7.3); extend the ledger to contracted and platform-mediated practitioners; let c_w depreciate under intermittent practice; and pilot the evidentiary record and contestability procedure of Section 5.2 against an existing municipal allocation practice, which a city could adopt tomorrow without the optimizer.

ABBREVIATIONS

AI, artificial intelligence; CAD, Civic Automation Debt; EU, European Union; GDPR, General Data Protection Regulation; HCPB, Human Capability Preservation Budget; ILO, International Labour Organization; ISO, International Organization for Standardization; LP, linear program; MDP, Markov decision process; SCV, Sustainable Civic Value.

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ETHICS STATEMENT

This study involved no human participants, no animal subjects, and no personal data. All parameter values reported are the authors' structured estimates, and all results are analytical. The expert-elicitation protocol specified in Appendix S5 has not been executed; its execution would require institutional ethics approval and written informed consent from participating practitioners, as Appendix S5.4 states. Ethical review and approval were therefore not required for this work in accordance with institutional and national requirements.

DATA AVAILABILITY STATEMENT

No new datasets were created or analyzed in this study. All parameter values, the worked allocation of Section 6, the sensitivity sweep of Section 6.8, the parameter shocks of Section 6.9, and the debt trajectory of Section 7.1 are fully specified in this article and can be reproduced from the reported values by direct arithmetic. The author-provided reference implementation, which recomputes every checkable number this article publishes and reports the agreement check by check, is available at <https://github.com/aliakarma/CivicWorkOS>. Code under `sim/` in that repository runs on clearly labeled synthetic demonstration data; it exercises the software shape of the protocol specified in Section 7.4 and Appendix S2, and it must not be cited as empirical municipal validation. The evaluation protocol of Section 7.4 has not been executed.

AUTHOR CONTRIBUTIONS

TAS and AAK conceptualized the study and developed the research methodology. RMA supervised the study and provided guidance on the research design and interpretation of the findings. MTN and AAI contributed to the investigation, formal analysis, and validation. TAS and AAK developed the reference implementation and its verification, and prepared the visualizations. TAS prepared the initial manuscript. All authors contributed to the revision and editing of the manuscript, critically reviewed the final version, and approved the submitted version. AAK denotes Ali Akarma and AAI Abdulaziz Alqurashi.

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CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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